



LAU

Lebanese American University

Chartered in the State of New York

www.lau.edu.lb

SOE

School of Engineering

*Department of Electrical &
Computer Engineering*

www.lau.edu.lb/ece



BYBLOS CAMPUS

P.O. Box: 36
Byblos, Lebanon
Tel: +961 9 547 262
Fax: +961 9 546 262

BEIRUT CAMPUS

P.O. Box: 13-5053 Chouran
Beirut 1102 2801, Lebanon
Tel: + 961 1 786 456
Fax: +961 1 867 098

NEW YORK HEADQUARTERS

211 E. 46th Street
New York, NY 10017-2935, USA
Tel: +1 212 203 4333
Fax: +1 212 784 6597

Intelligent Data Processing and Applications

COE 543/743

Chapter 10 - Data Clustering

Joe M. Tekli, PhD

Computer Software Engineering

Ch. 10 – Data Clustering

Chapter Contents

1. Introduction
2. Machine Learning Paradigms
3. Notion of Cluster
4. Ingredients of Clustering
5. Clustering Algorithms
 - 5.1 Partitional
 - 5.2 Hierarchical
 - 5.3 Others
6. Clustering Evaluation

Ch. 10 – Data Clustering

1. Introduction

- **What is clustering?**
 - Organizing data into so-called *clusters* such that there is:
 - **High** intra-cluster similarity
 - **Low** inter-cluster similarity
- In other words, finding **groups of data** objects such that
 - Objects in a group will be **similar** (or related) to one another
 - And **different** from (or unrelated to) objects from other groups

Informally

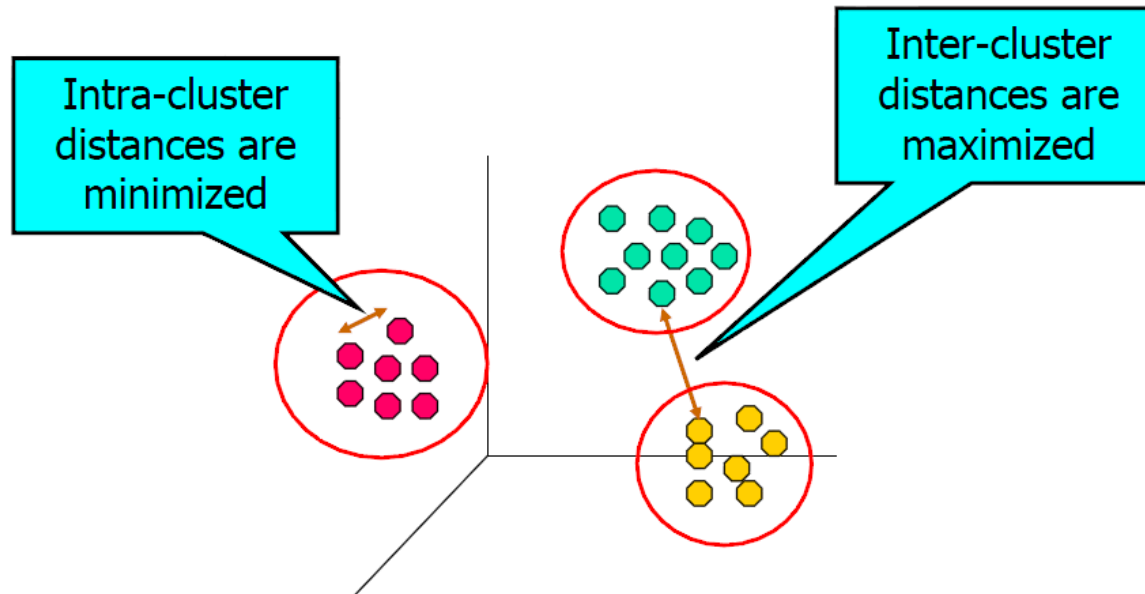


Finding natural groupings among objects

Ch. 10 – Data Clustering

1. Introduction

- What is clustering?



Why would we want to cluster?
Any real applications?

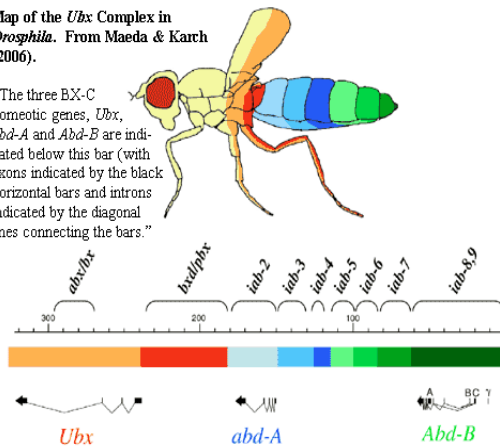
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1. Introduction

- Clustering is useful in many application scenarios
 - Pattern detection:** reflecting internal and **hidden structure** of data
 - E.g., **Clustering genes:** studying their properties in different conditions

Map of the *Ubx* Complex in *Drosophila*. From Maeda & Karch (2006).

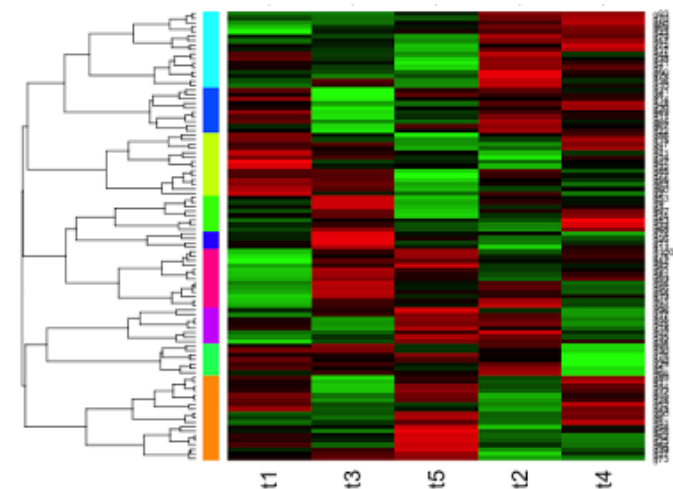
"The three BX-C homeotic genes, *Ubx*, *abd-A* and *Abd-B* are indicated below this bar (with exons indicated by the black horizontal bars and introns indicated by the diagonal lines connecting the bars.)"



Acquiring genes



Describing genes



Clustering genes

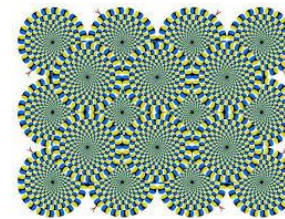
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1. Introduction

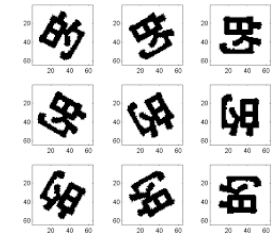
- Clustering is useful in many application scenarios

1. Pattern detection: reflecting internal and **hidden structure** of data

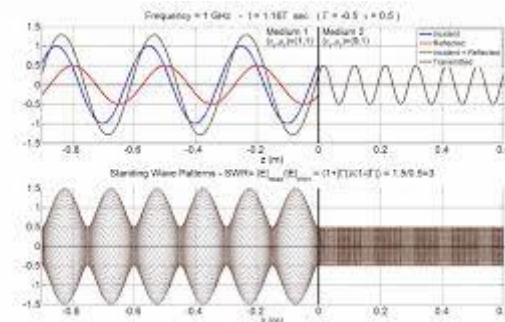
- A pattern can be
 - A textual expression
 - An audio signal
 - A still image
 - A moving image
 - A radio-wave
 - A sound-wave
 - A part of speech
 - ...



Repeating visual patterns



Hand writing patterns



Ratio wave patterns



Star configuration patterns

Ch. 10 – Data Clustering

1. Introduction

- Clustering is useful in many application scenarios

2. Knowledge discovery: topics, relationships, connections between data

- Useful in **data search**, **organization**, and **integration**

Example:
Clusty search engine

Grouping results
related to search
request in
coherent topics

The screenshot shows the LessonPaths interface with a search for 'Lebanon' using the Clusty search engine. The results are grouped into coherent topics, as indicated by the callout box. The interface includes a search bar, a 'NEXT' button, and a list of search results.

Search Engines **NEXT** **4 Steps Left**

yippy Lebanon

Sources Sites Time Topics

- + Turkey, Syria (15)
- + Lebanon, Pennsylvania (19)
- + Saudi Arabia (20)
- + Dealer (16)
- + Lebanon, Ohio (21)
- + Church (21)
- + Istanbul, Deadly (10)
- + Lebanon, New Hampshire (16)
- + Lebanon, Missouri (13)
- + Mount Lebanon (12)
- + Liberty (14)

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[lebanon.com](#) - [cache](#) - Yippy Index

Lebanon Illinois Directory Page [new window](#) [preview](#)
Lebanon Chamber of Commerce lebanonil.us **Lebanon Community** Unit School District #9 lcusd9.org Mc magic in **Lebanon**. Annual Events: Chocolate Rendevous for Valentine's Day, ... a residence, this building w Asphalt is one of ...
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... INFO ABOUT US YOUR INTERNET GUIDE TO **LEBANON** **LEBANON** DISCUSSION FORUMS General C Politics, Sex & Religion in **Lebanon** - Have fun... if you can. Politics in the Middle-East - How do you like p food? Share your secrets. ...

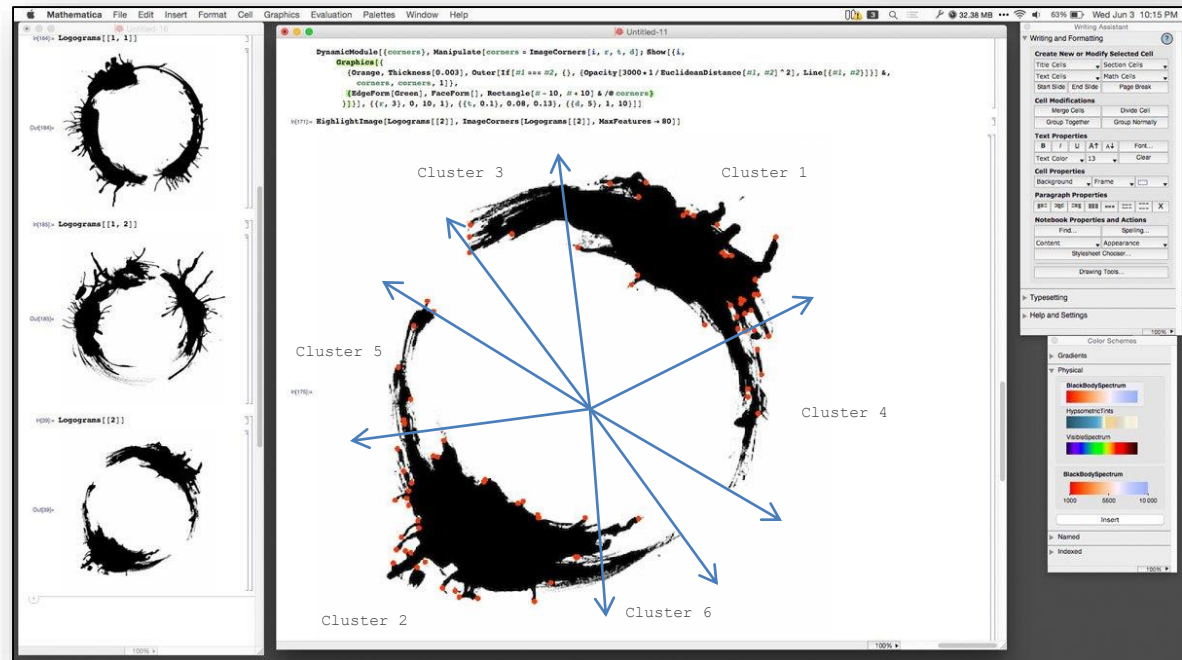
Ch. 10 – Data Clustering

1. Introduction

- Clustering is useful in many application scenarios

3. Pre-processing: Preparing for other AI techniques

- Useful for **pattern recognition**: e.g., deciphering alien alphabet



Ch. 10 – Data Clustering

1. Introduction

- Clustering is useful in many application scenarios

3. Pre-processing: Preparing for other AI or computation techniques

- Useful in NLP, e.g., **text summarization, indexing, and semantic disambiguation**



Emails



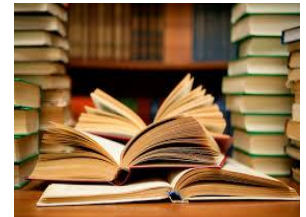
News articles



Scientific papers



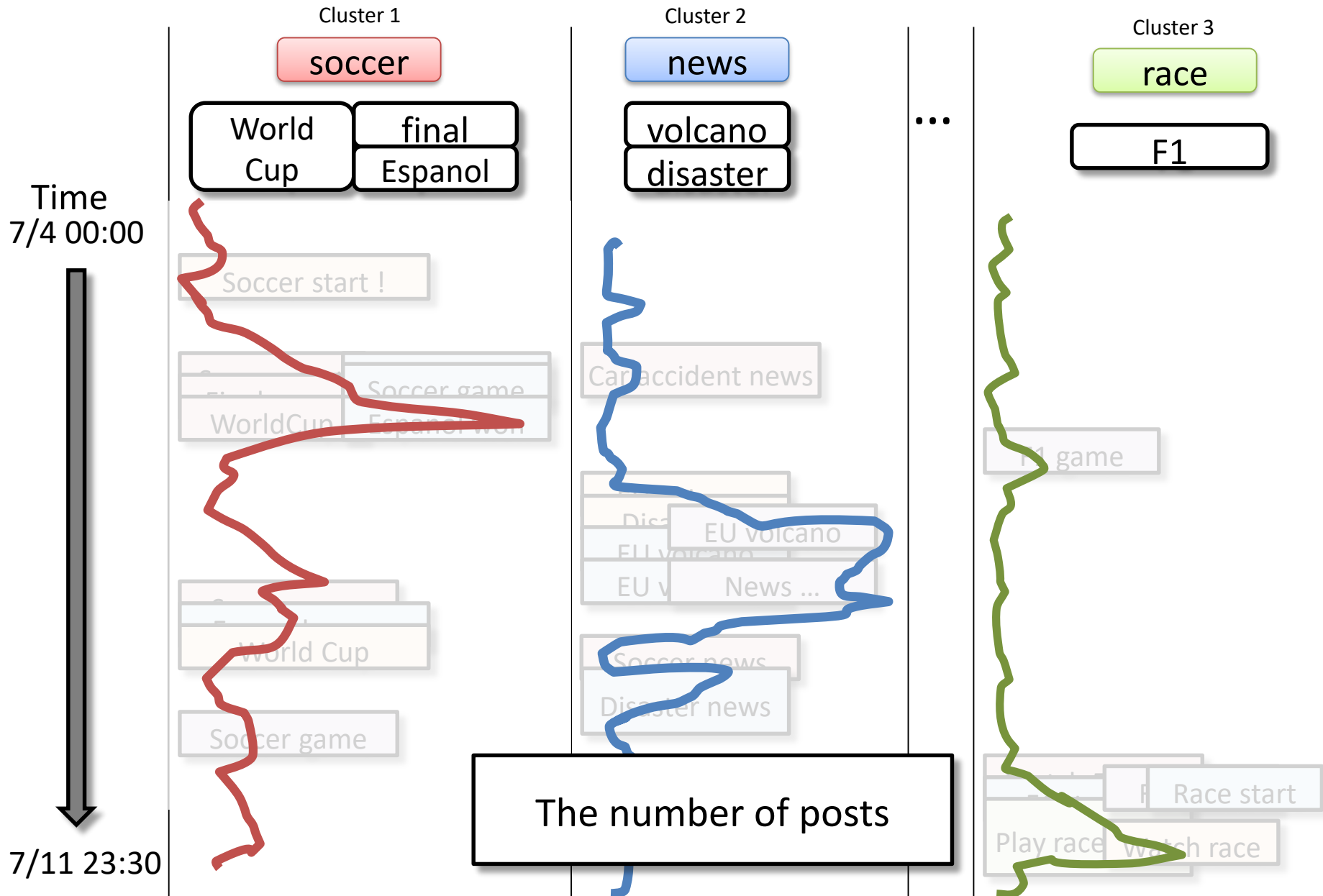
Websites



Books

- **Mining social media:** Facebook, Twitter, Instagram, You Tube, etc.
 - For commercial, business, and security purposes

Example: Twitter message cluster-based mining



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2. Machine Learning Paradigms

- **Machine learning** is the science of getting computers to act without being explicitly programmed
- In the past decade, **machine learning** has given us:
 - Self-driving cars, practical speech recognition, effective web search, among other applications, as well as a vastly improved understanding of the human genome...



Key to making progress towards human-level AI

- Machine learning techniques fall within two main categories:
 - **Supervised** learning
 - **Unsupervised** learning

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2. Machine Learning Paradigms

2.1. Supervised Learning

- **Main premise:** Identifying an **output measure** based on **input data**
 - By mapping inputs to **desired output**
 - Using sample input/output data provided by experts: **training data**
- Training data provided in the form of examples $(\mathbf{X}_1, \mathbf{y}_1), \dots, (\mathbf{X}_m, \mathbf{y}_m)$
 - where $\mathbf{X}$ is a vector of $\mathbf{x}_1, \dots, \mathbf{x}_k$ input values
 - And $\mathbf{y}$ is the output generate by **unknown/desired** activation function f



Discover a function h that approximates the desired activation function f

- Approach known as **classification**

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2. Machine Learning Paradigms

2.1. Supervised Learning

(1) *Data collection*



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2. Machine Learning Paradigms

2.1. Supervised Learning

(1) Data collection



(2) Known categories

Cat. 1



Cat. 2



Cat. 3



Cat. 4



Cat. 5



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2. Machine Learning Paradigms

2.1. Supervised Learning

(1) Data collection



(2) Known categories



(3) Compare
with
category
descriptions

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2. Machine Learning Paradigms

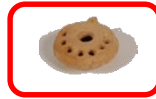
2.1. Supervised Learning

(1) Data collection



(2) Known categories

Cat. 1



Cat. 2



Cat. 3



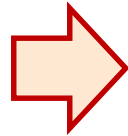
Cat. 4



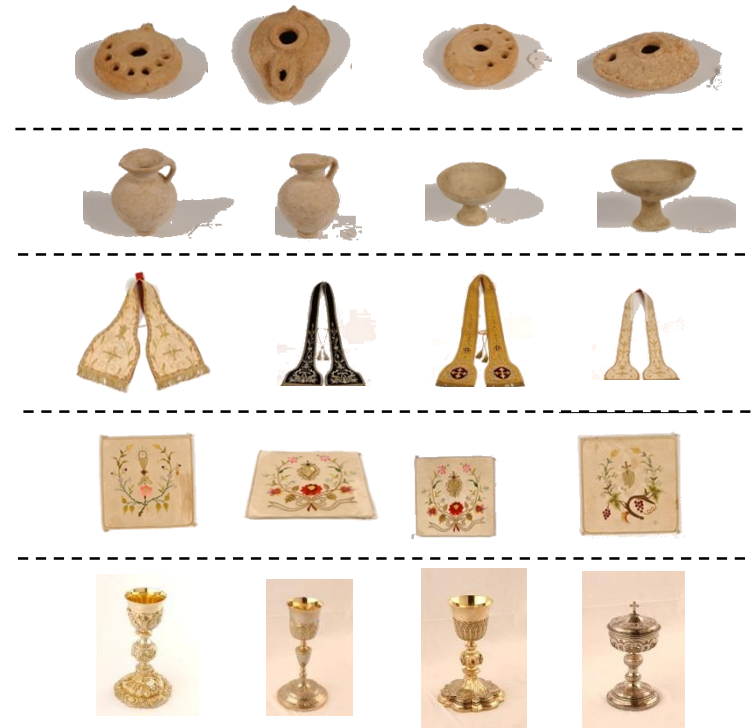
Cat. 5



(3) Compare
with
category
descriptions



(4) Identifying classes of objects



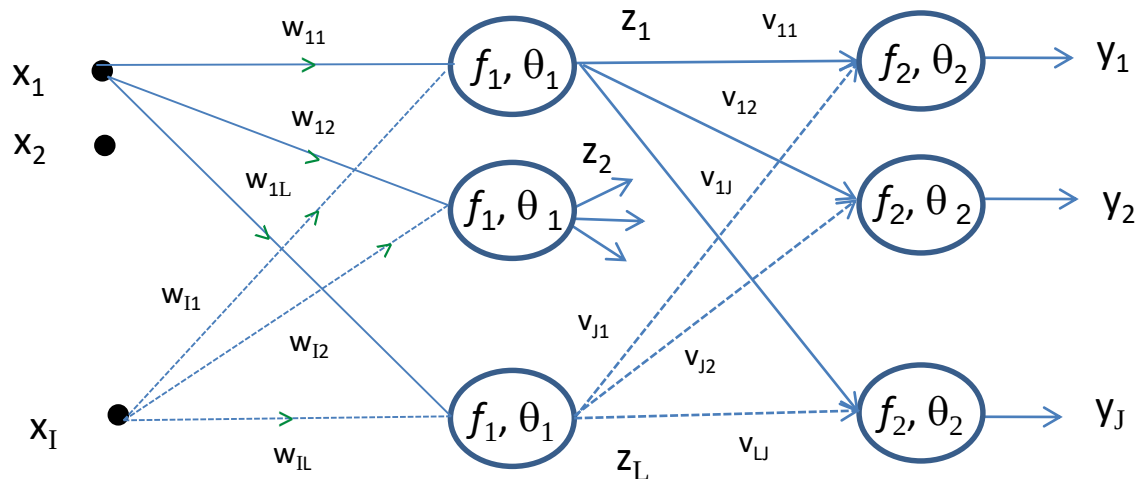
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2. Machine Learning Paradigms

2.1. Supervised Learning

– Different **classification algorithms** exist in the literature

- E.g., Instance-based, Support Vector Machines, and **Artificial Neural Networks**



Two-layer
feed-forward
neural network,
known as
perceptron



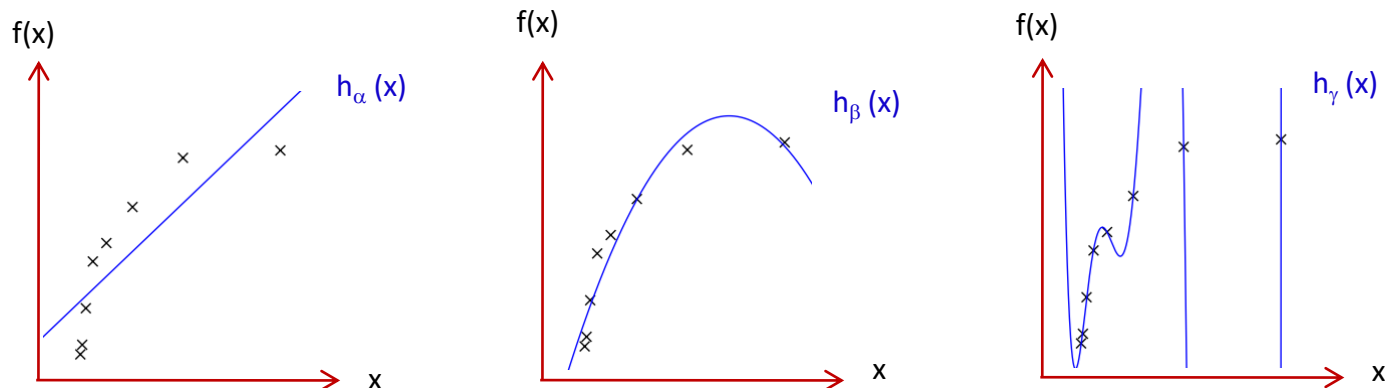
Inspired by **biological brain**: ideal **learning/recognition** system

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2. Machine Learning Paradigms

2.1. Supervised Learning

- When the output of classifier is a numerical value (e.g., temperature degree)
 - The learning problem is called **regression analysis**



- We can approximate f with different hypotheses, e.g., h_α , h_β , and h_γ
 - $\Rightarrow$ We usually choose the **simplest** (following **Ockham's razor**)
 - Major category of techniques allowing predictability

* Methodological principle by W. Ockham (1287-1347): "The simplest explanation tends to be the right one"

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2. Machine Learning Paradigms

2.2. Unsupervised Learning

- **Main premise:** Finding a **hidden pattern**/structure in unlabeled data
 - Absence of an expert/knowledge about expected outputs for inputs
 - ⇒ **Lack of training data:** Learning **without** external **intervention**
- Typical approach: **clustering**
- Other approaches:
 - **Feature extraction** techniques
 - Identifying most important features in describing the data
 - ⇒ E.g., identifying most important terms in describing (a) document(s)
 - **Dimension transformation and reduction** techniques
 - Such as: Principal Component Analysis (PCA) and Initial Component Analysis (ICA)
 - Useful in textual data processing and retrieval

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2. Machine Learning Paradigms

2.2. Unsupervised Learning

Example: Web Image Clustering



(1) Input unlabeled data

*Without any
previous knowledge
about data organization*

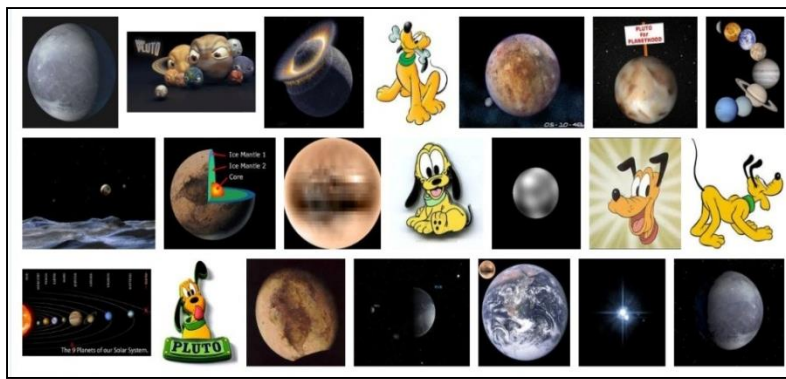
⇒ No predefined categories

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2. Machine Learning Paradigms

2.2. Unsupervised Learning

Example: Web Image Clustering



(1) Input unlabeled data

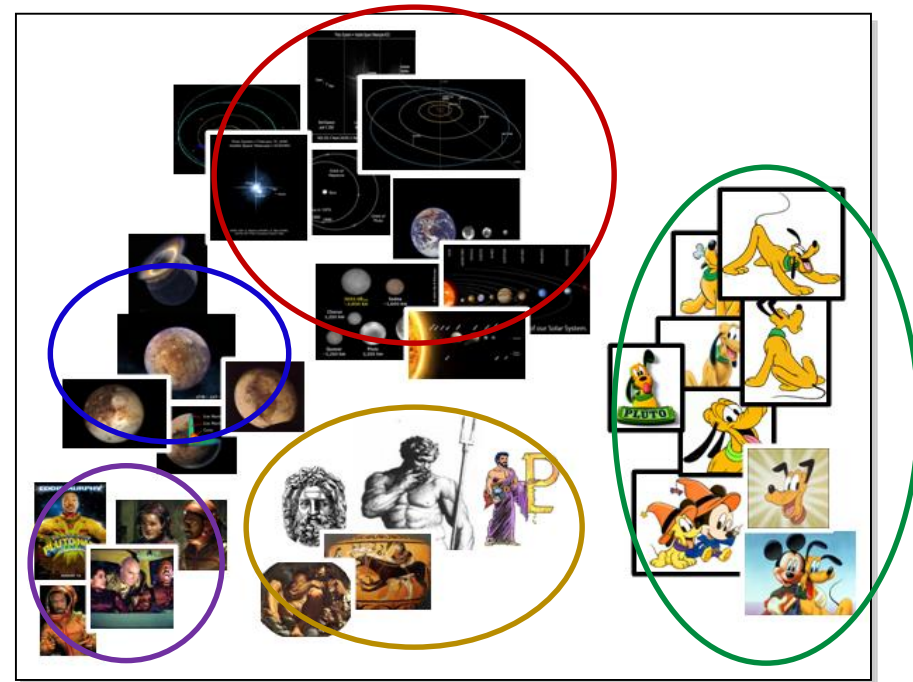
*Without any
previous knowledge
about data organization*

⇒ No predefined categories



(2)

Comparing
images
against
each other



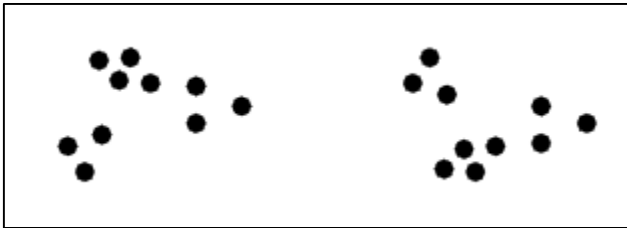
(3) Data clustered in 5 groups

⇒ Identifying 5 clusters of similar images

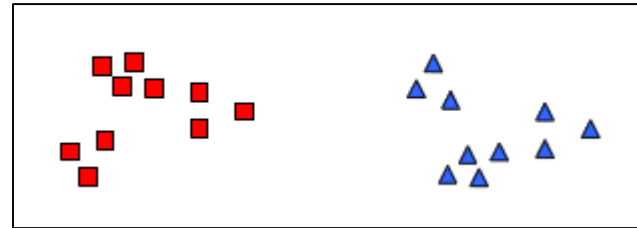
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3. Notion of Cluster

- The notion of cluster can be ambiguous
 - Consider the following dots representing objects in a 2D space



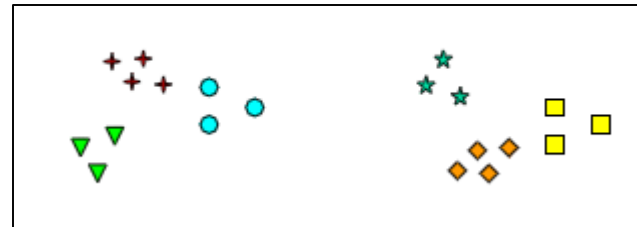
How many clusters can we identify?



Two clusters?



Four clusters?



Maybe six clusters?

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3. Notion of Cluster

- Clusters can be of different types:
 - **Well separated**
 - **Center based**
 - **Contiguous**
 - **Density based**
 - **Property based** (or conceptual)



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3. Notion of Cluster

3.1. Well Separated Clusters

- A cluster is a set of objects such that **any object** in the **cluster** is **closer** (more similar) **to every other object** in the **same cluster**
 - **Than to** any other **object outside** of the cluster



Cluster 1



Cluster 2



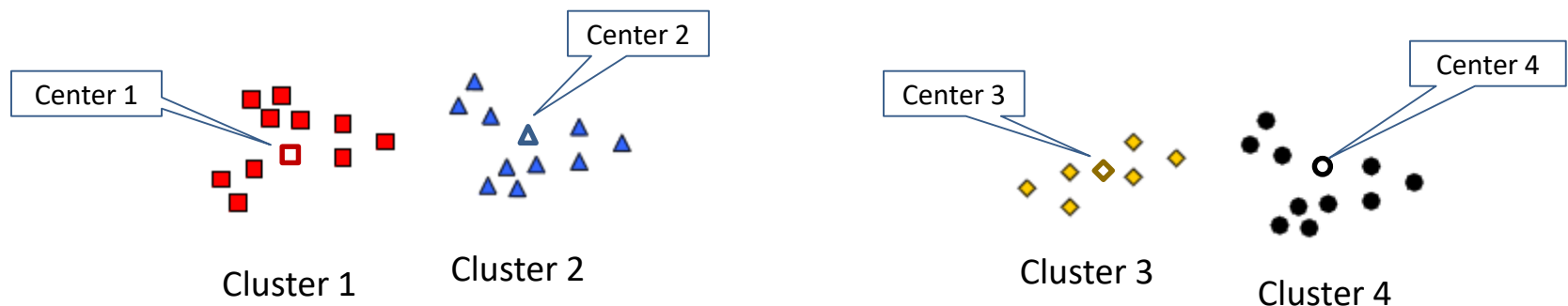
Cluster 3

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3. Notion of Cluster

3.2. Center-based Clusters

- A cluster is a set of objects such that **any object in the cluster is closer** (more similar) **to the “center” of the cluster**
 - Than to the **center of every other cluster**
 - The center of a cluster can be:
 - A **centroid**: average of all objects in the cluster
 - A **medoid**: the most “representative” object in the cluster

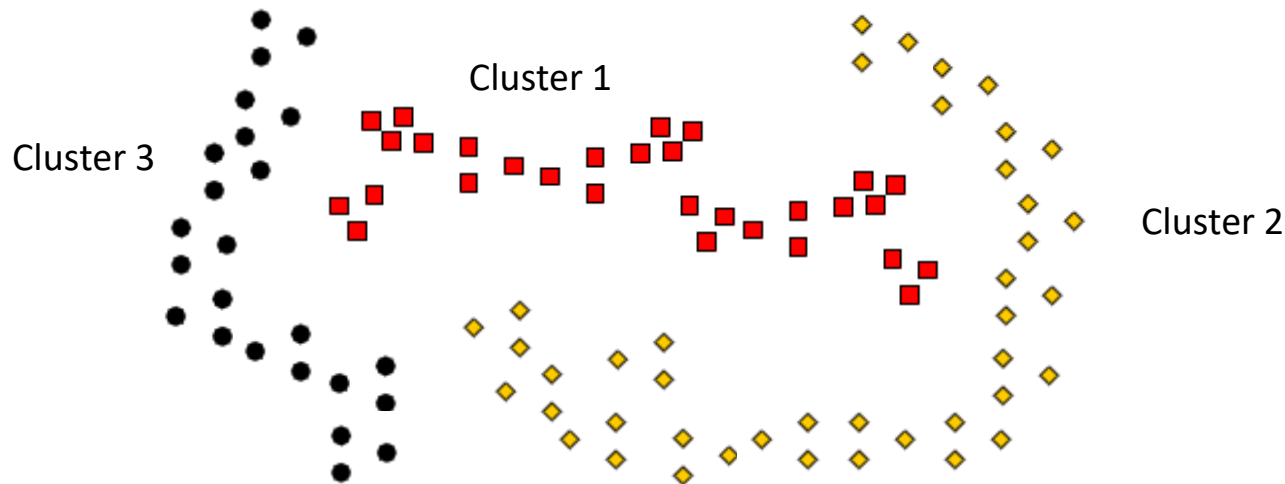


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3. Notion of Cluster

3.3. Contiguous Clusters

- A cluster is a set of objects such that **any object in the cluster is closer** (more similar) **to one or more** other **objects** in the cluster
 - Than to any **object outside** of the cluster

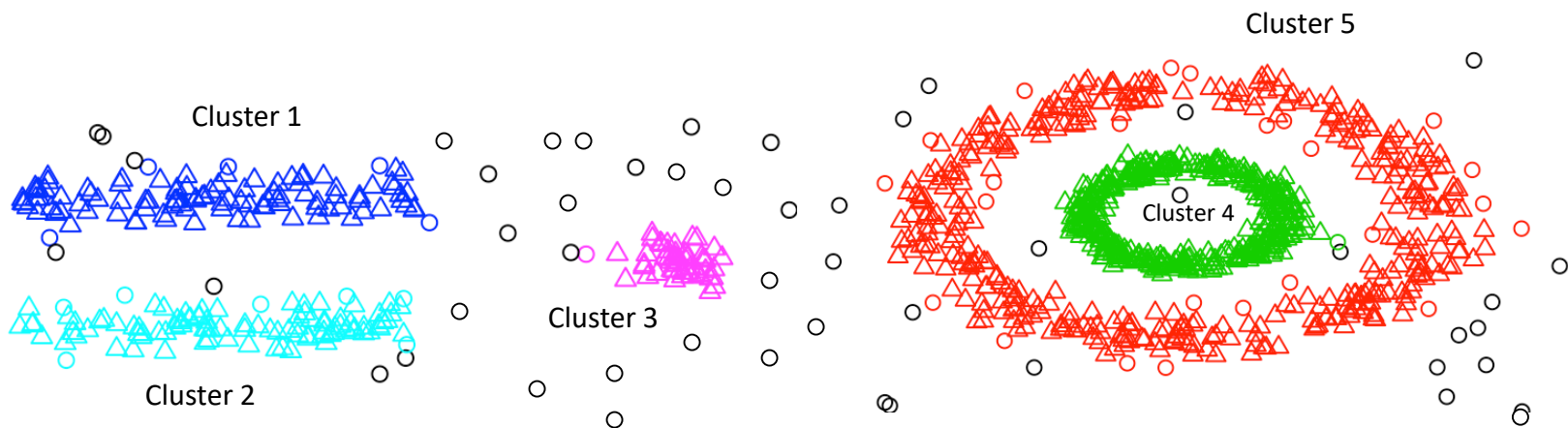


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3. Notion of Cluster

3.4. Density-based Clusters

- A cluster is a **dense region of objects**, **separated by low density regions**, from other regions of high density
 - Useful when clusters are **irregular** or **intertwined**
 - And when **noise** and **outliers** are present

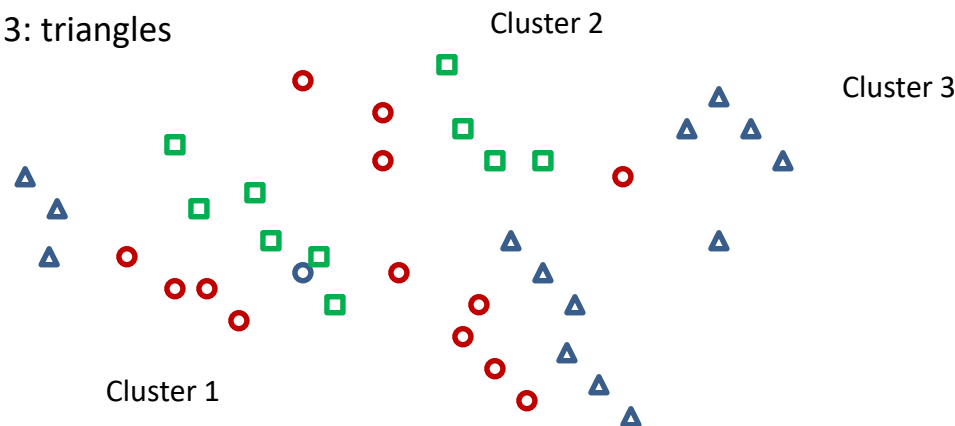


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3. Notion of Cluster

3.5. Shared Property Clusters

- Or conceptual clusters: clusters that share **common or similar properties** to describe a certain concept
 - Example: grouping objects based on *geometric shape* property
 - Cluster 1: circles
 - Cluster 2: squares
 - Cluster 3: triangles



Ch. 10 – Data Clustering

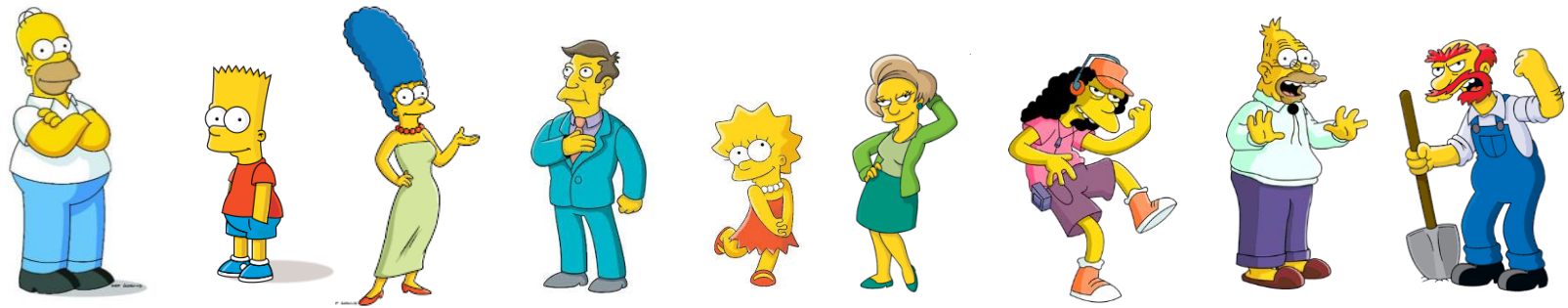
4. Ingredients of Clustering

- Clustering requires two basic ingredients
 - **Similarity (distance) computation**
 - Necessary to compare data objects being clustered
 - **Clustering algorithm**
 - Algorithmic process to perform the clustering task
- Clustering also requires a **pre-processing** phase
 - **Feature selection, extraction, and normalization**
 - Based on which similarity computation will be performed
 - E.g., with XML: ROLT representation, LD-pair sequence, content vector, etc.
 - E.g. with images: dominant color, color histogram, shape histogram, surrounding text, etc.

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4. Ingredients of Clustering

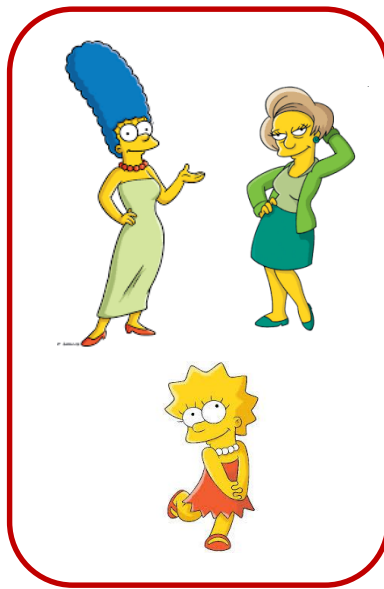
- For instance: what is a natural grouping of these objects?



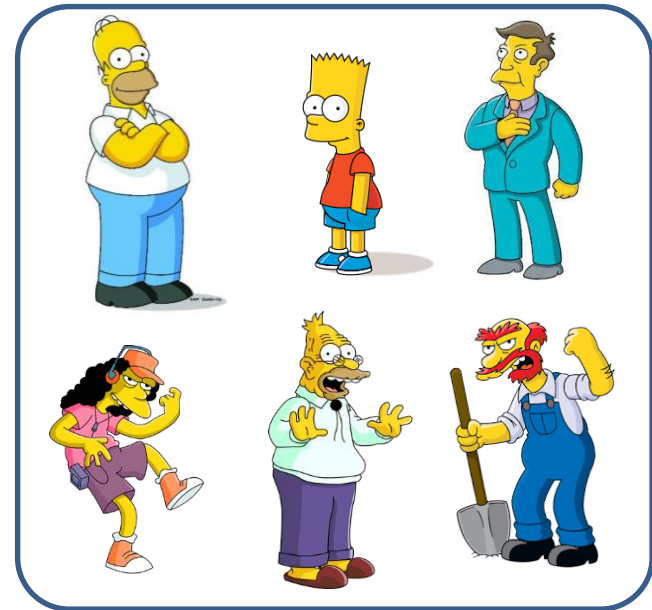
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4. Ingredients of Clustering

- It is first and fore-most **feature-dependent**:
 - Clustering based on **gender** feature:



Cluster 1

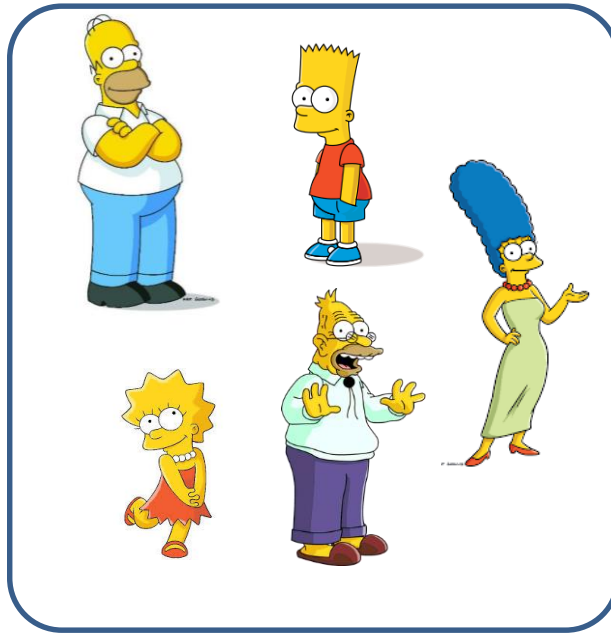


Cluster 2

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4. Ingredients of Clustering

- It is first and foremost **feature-dependent**:
 - Clustering based on **family ties** feature:



Cluster 1



Cluster 2

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4. Ingredients of Clustering

- Actually, **similarity** is **feature dependent**:
 - And it affects clustering accordingly:



- It not only depends on the **similarity measure** that we adopt
 - But most importantly: the **features** that we target in our similarity computation

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4. Ingredients of Clustering

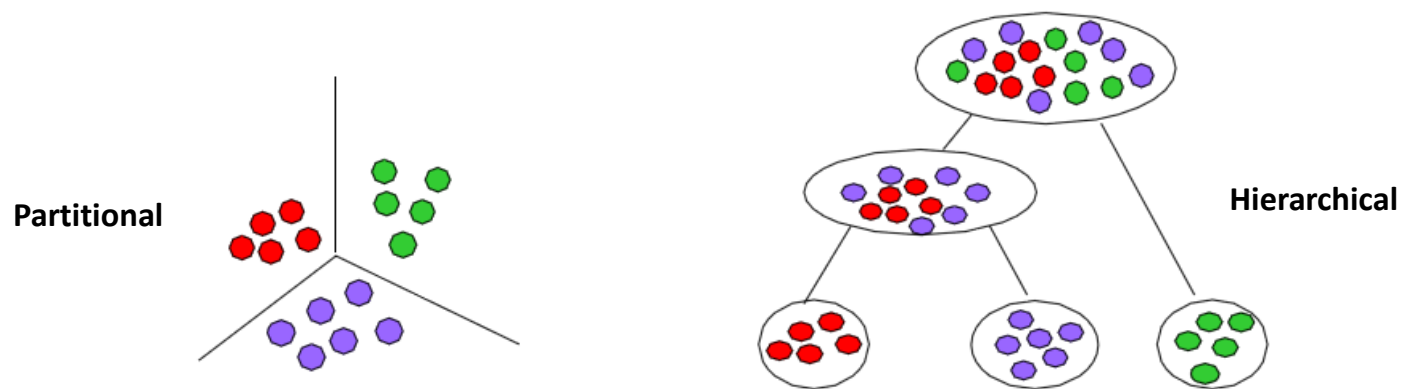
- NOTE: It is advised to adopt the **formal definition of similarity**
 - **Inverse** of a metric **distance** function
 - For the clustering to work properly: to be (partly) traceable/verifiable
- **Similarity metric**: function verifying the **metric properties**, namely:
 - $\text{Sim}(O_1, O_2) \leq \text{Sim}(O_1, O_1) \Rightarrow$ **Minimality**
 - $\text{Sim}(O_1, O_2) = \text{Sim}(O_2, O_1) \Rightarrow$ **Symetricity**
 - $\text{Sim}(O_1, O_3) \geq \text{Sim}(O_1, O_2) \times \text{Sim}(O_2, O_3) \Rightarrow$ **Triangular inequality**
 - $\text{Dist}(O_1, O_3) \leq \text{Dist}(O_1, O_2) + \text{Dist}(O_2, O_3)$



Ch. 10 – Data Clustering

5. Clustering Algorithms

- Different clustering strategies exist, categorized in three main groups:
 - **Partitional:** dividing of data objects into non-overlapping sub-sets, i.e., the clusters, such that each data object is in exactly one cluster
 - **Hierarchical:** generating a set of nested clusters organized in a hierarchy of clusters, called *dendrogram*
 - **Others:** different hybrid techniques



Ch. 10 – Data Clustering

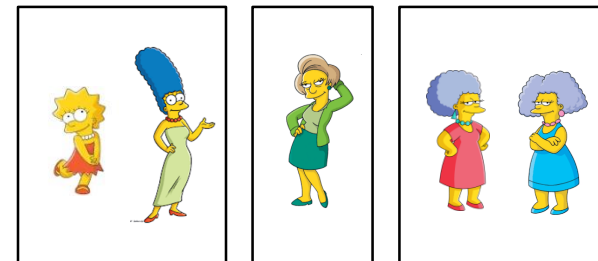
5. Clustering Algorithms

How to choose adapted clustering algorithm?

- Based on the user's desirable properties
 - **Scalability** (in terms of both time and space)
 - Ability to deal with **different data types**
 - **Minimal requirements** for domain knowledge to determine **input parameters**
 - **Insensitive to order** of input records
 - **Interpretability** and **usability**
 - Able to deal with **noise** and **outliers**
 - Incorporation of **user-specified constraints**

5. Clustering Algorithms

- **Partitional clustering:** dividing of data objects into non-overlapping sub-sets, i.e., the clusters, such that each data object is in exactly one cluster
 - Also known as **non-hierarchical** algorithms



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5. Clustering Algorithms

5.1. Partitional Clustering: K-means

- **K-means** is one of the most popular partitional clustering algorithms
 - Requires the number of clusters **k** to be specified by the user
- Main process:

 1. Initialization: Select **k** data objects (points) as the initial **centroids**
 2. **Repeat**
 3. Assignment: Form **k** clusters by assigning objects to the **closest centroids**
 4. Update: Re-compute the **centroid** for each cluster
 5. **Until** reaching **convergence**

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5. Clustering Algorithms

5.1. Partitional Clustering: K-means

- **Centroid:** computed as the **mean** of all data objects in the cluster
 - **Initial centroids** are chosen **randomly**: typically among the available data objects
- **Closeness to centroid:** evaluated using a **similarity (distance) measure**
 - E.g., using XML similarity when clustering XML documents
- **Convergence:** is reached when few (or no) data objects **change clusters**
 - Between the previous iteration and the current one

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.1. Partitional Clustering: K-means

- Mathematically:
 - Initialization: Given **k** chosen by user
 - Randomly choose initial **k** centroids (means) at iteration **t₀**: $m_1^{(t_0)}, \dots, m_k^{(t_0)}$
 - Assignment: Given the **k** centroids at iteration **t**
 - Compare every object **x_p** with every centroid **m_i**
 - And assign **x_p** to cluster **C_i** corresponding to most similar centroid **m_i**
- Update: Re-computing centroids as means of data objects in every cluster
 - To be used as new centroids in the following iteration

$$\Rightarrow m_i^{(t+1)} = \frac{1}{|C_i^{(t)}|} \sum_{x_p \in C_i^{(t)}} x_p$$

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.1. Partitional Clustering: K-means

- Mathematically:
 - Convergence: Can be evaluated in different ways:
 1. Verifying that **no (or few) object(s)** in iteration **t** have changed clusters in iteration **t+1**
 - The user can choose the maximum number of changes allowed: **threshold number**
 2. Verifying that clusters at iteration **t** have **higher (or close enough) intra-cluster similarity** than clusters at iteration **t+1**
 - Usually evaluated as **Sum of Squared Error** (or SSE)

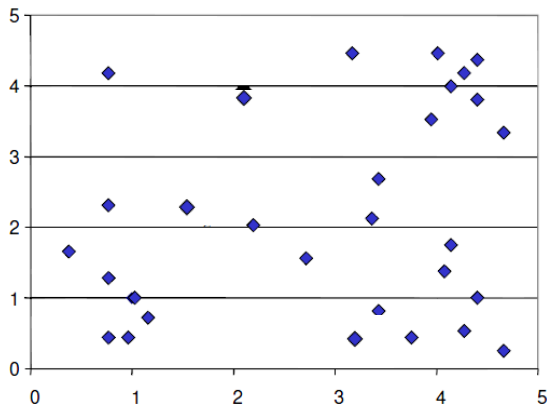
$$\Rightarrow SSE^{(t)} = \sum_{i=1}^k \sum_{x_p \in C_i^{(t)}} Dist(x_p, m_i^{(t)})^2 \quad \text{where } Dist \text{ is the inverse function of } Sim$$
 - $\Rightarrow$ **Higher intra-cluster similarity** means **lower SSE**
 - User can also choose allowed SSE difference threshold

Ch. 10 – Data Clustering

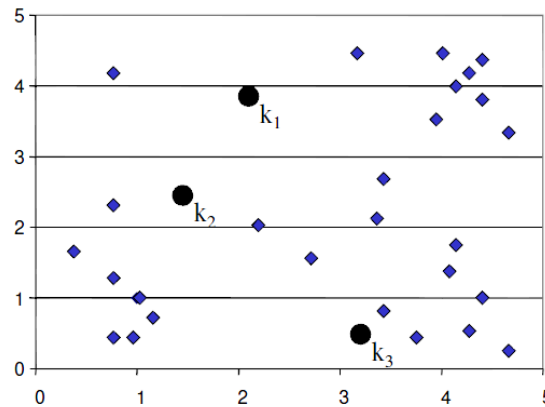
5. Clustering Algorithms

5.1. Partitional Clustering: K-means

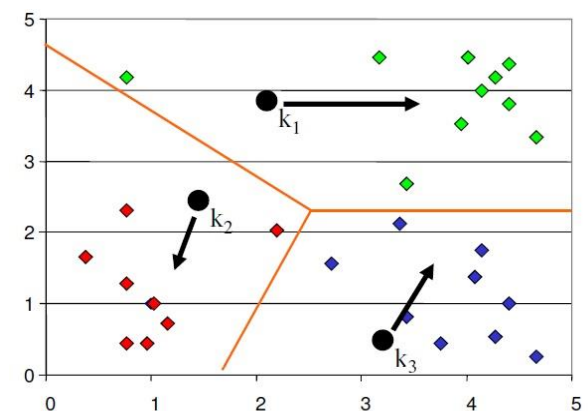
- Example: K-means clustering
 - Clustering a set of data objects represented as points in a 2D space
 - The user chooses $k = 3$, i.e., needs to produce three clusters



(1) Set of points



(2) **Initialize** k centroids (randomly)



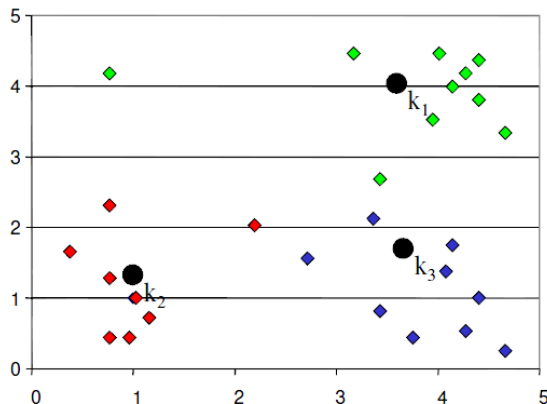
(3) **Assign** points to the nearest centroid, then **update** (compute new) centroids

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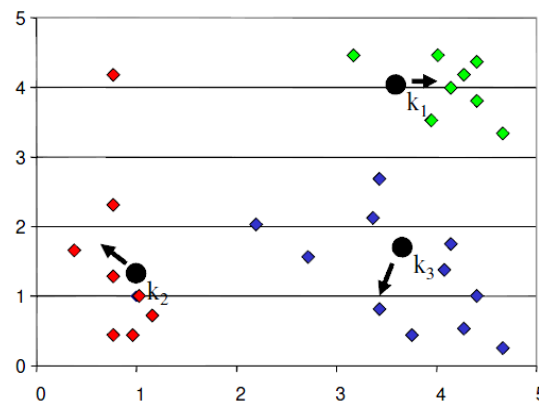
5. Clustering Algorithms

5.1. Partitional Clustering: K-means

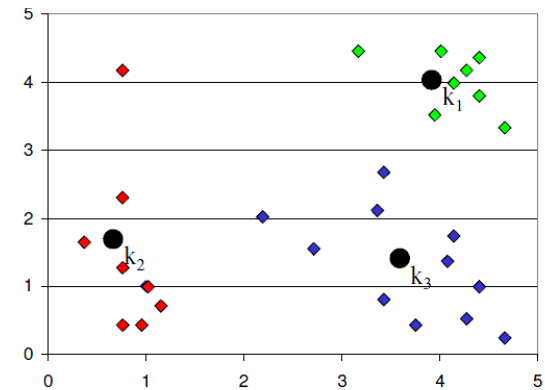
- Example: K-means clustering (cont'd 2)
 - Start of Iteration 2:



(4) After updating centroids, re-assign points to new centroids



(5) The **update** (compute new) centroids



(5) No objects changed membership
⇒ **Convergence** reached

DONE!

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.1. Partitional Clustering: K-means

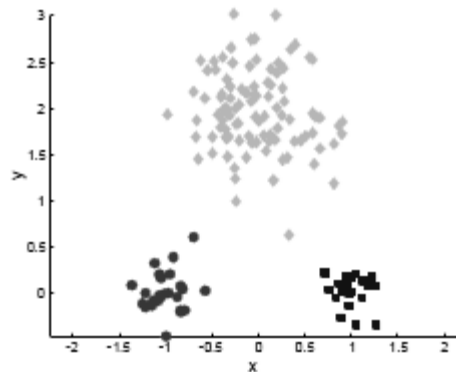
- **Pros** of k-means:
 - **Intuitive** objective function: optimizing intra-cluster similarity
 - **Easy to implement** and debug
 - Relatively **efficient**: of average $O(n \times k \times i)$ time and $O(n)$ space complexity, where **n** is # of objects, **k** is # of clusters, and **i** is # of iterations
- **Cons** of k-means:
 - Need to **specify k**, the number of clusters, in advance
 - Applicable only when mean can be defined: what about **categorical (or set) data**?
 - Results **depend on** the **initial centroids**: often terminating at **local optima**
 - **Not suitable** to identify **clusters with different sizes and densities**
 - Also has problems when clusters have **non-convex** (non-globular) **shapes**
 - **Unable** to handle **noise** or **outliers**

Ch. 10 – Data Clustering

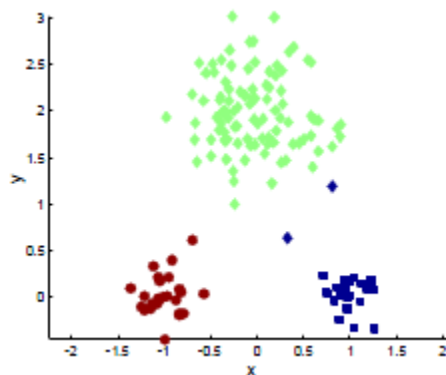
5. Clustering Algorithms

5.1. Partitional Clustering: K-means

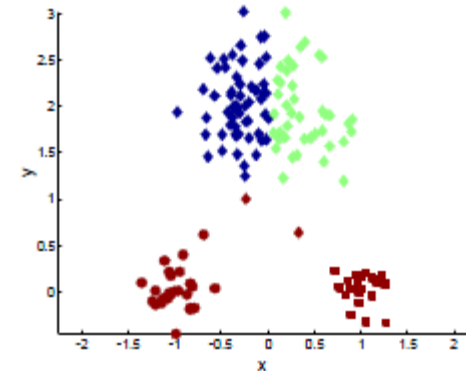
- Discussion: Dependency on **initial centroids**
 - Consider the following set of points
 - Different possible clustering solutions can be obtained with every run of k-means



Original set of points
(shapes of points represent desired clusters)



Possible solution 1: Optimal

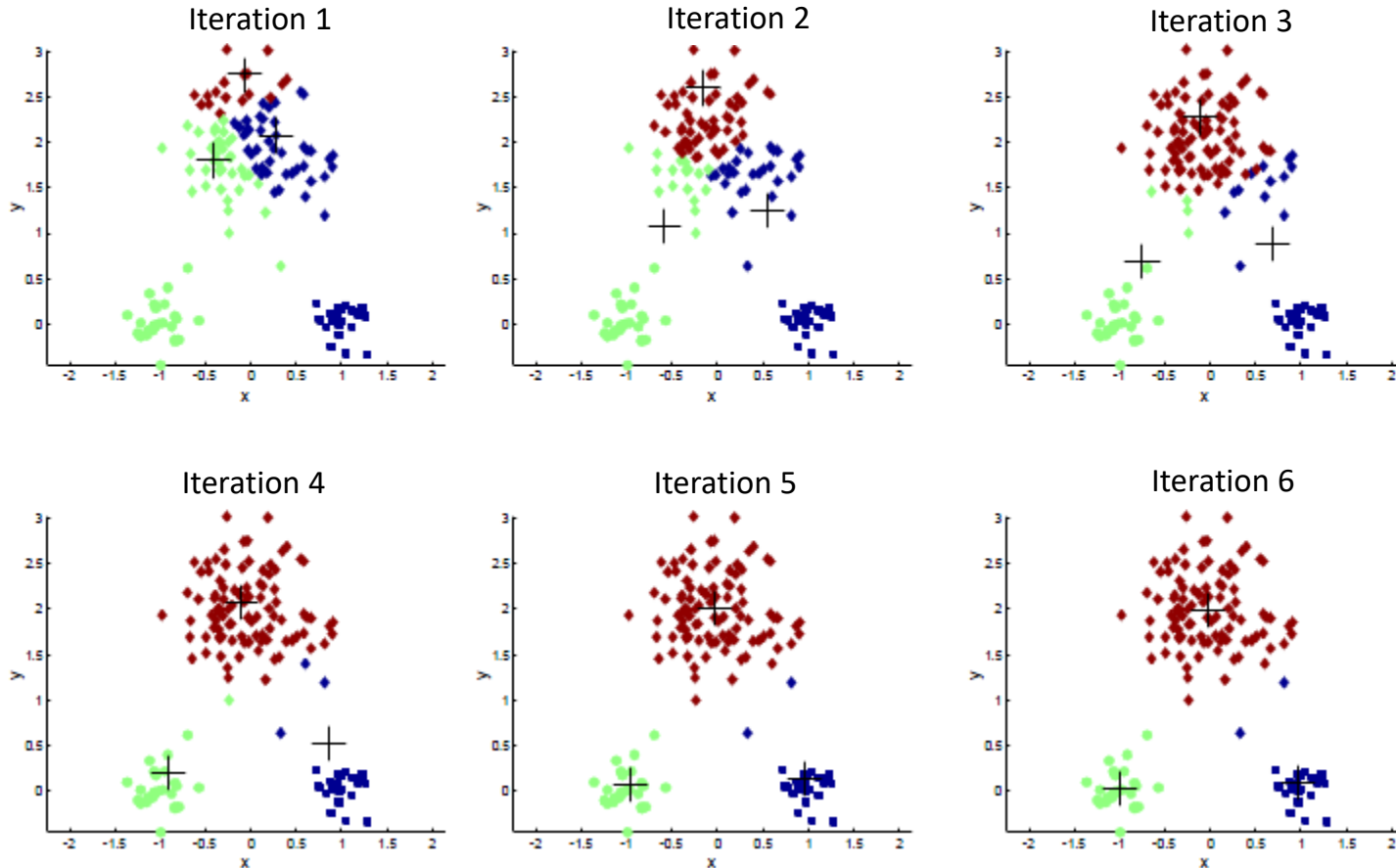


Possible solution 2: Sub-Optimal

5. Clustering Algorithms

5.1. Partitional Clustering: K-means

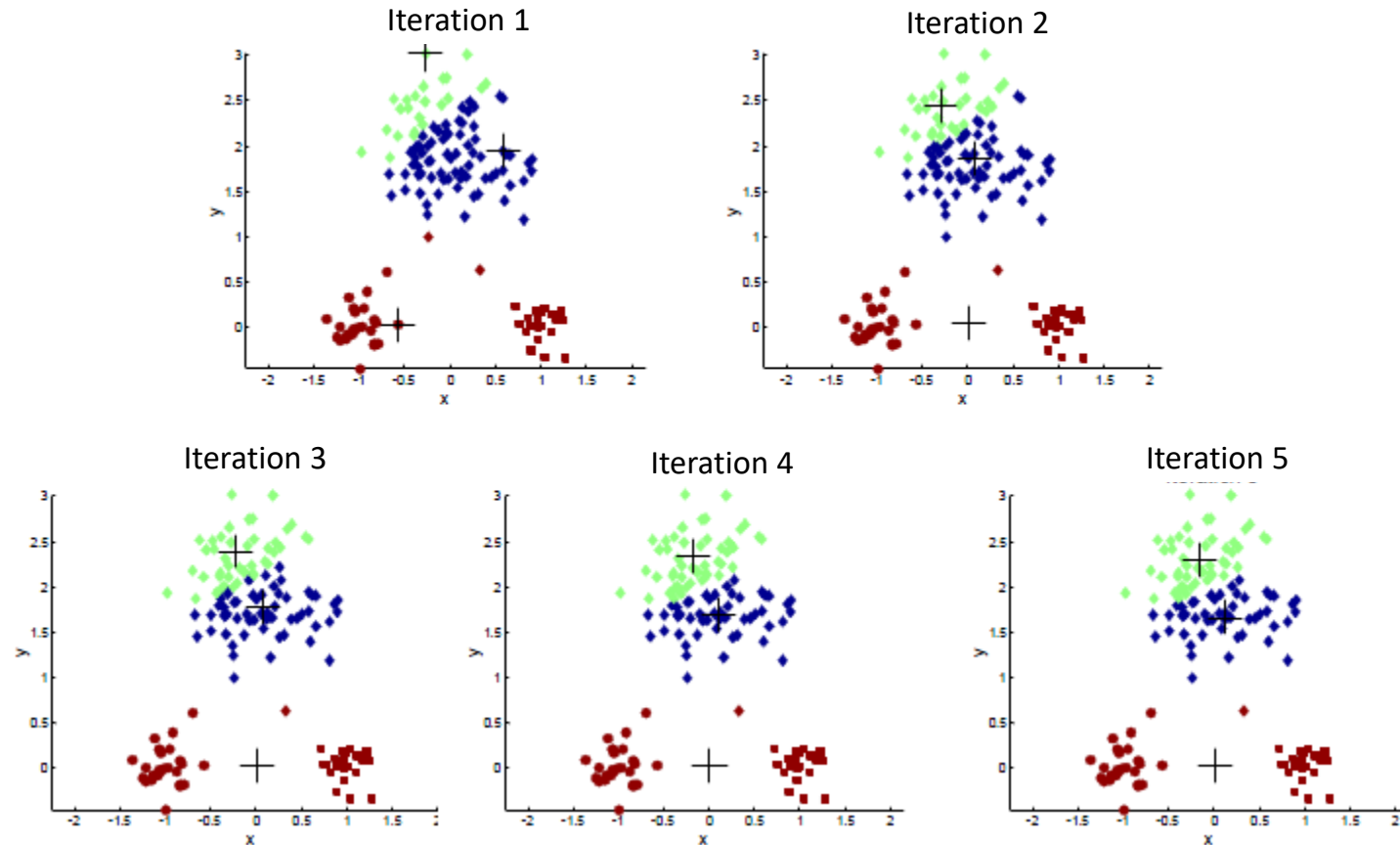
- Discussion: Dependency on **initial centroids** - *Possible solution 1*



5. Clustering Algorithms

5.1. Partitional Clustering: K-means

- Discussion: Dependency on **initial centroids** - *Possible solution 2*

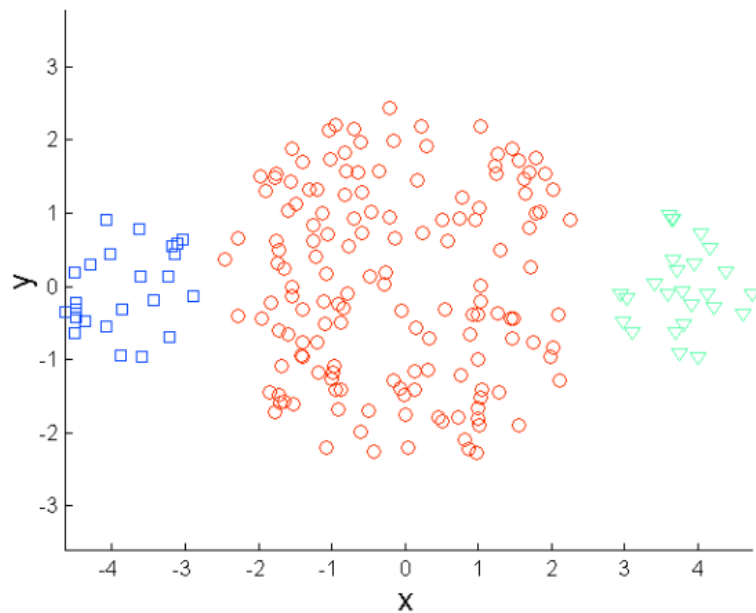


Ch. 10 – Data Clustering

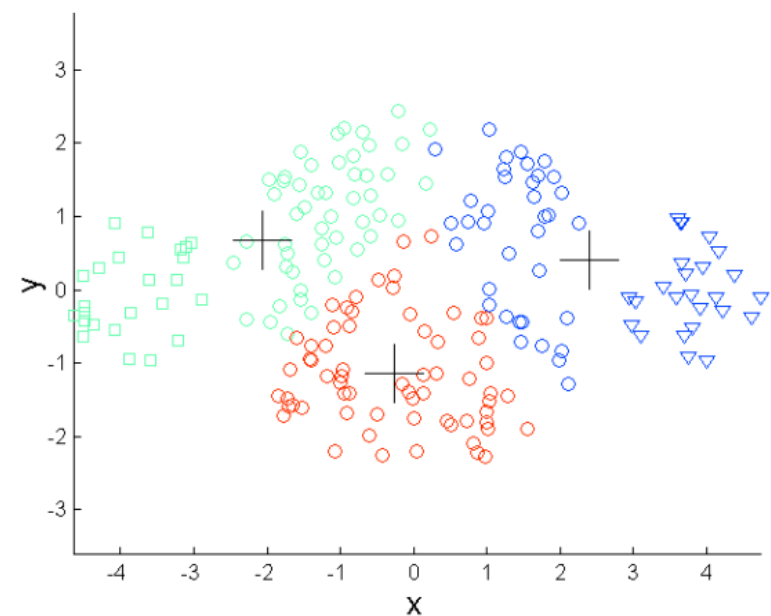
5. Clustering Algorithms

5.1. Partitional Clustering: K-means

- Discussion: Difficult to produce clusters with **different sizes**



Original set of points
(colors of points represent desired clusters)



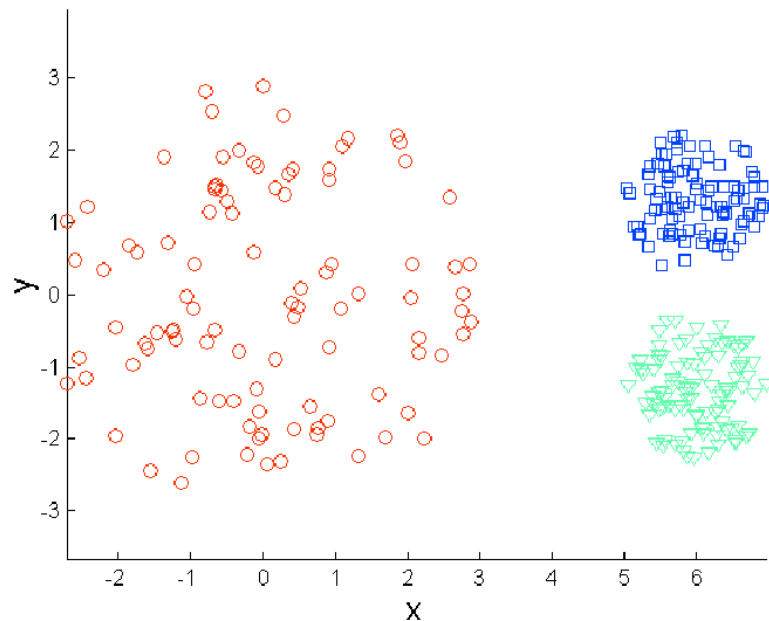
Expected k-means result with k=3

Ch. 10 – Data Clustering

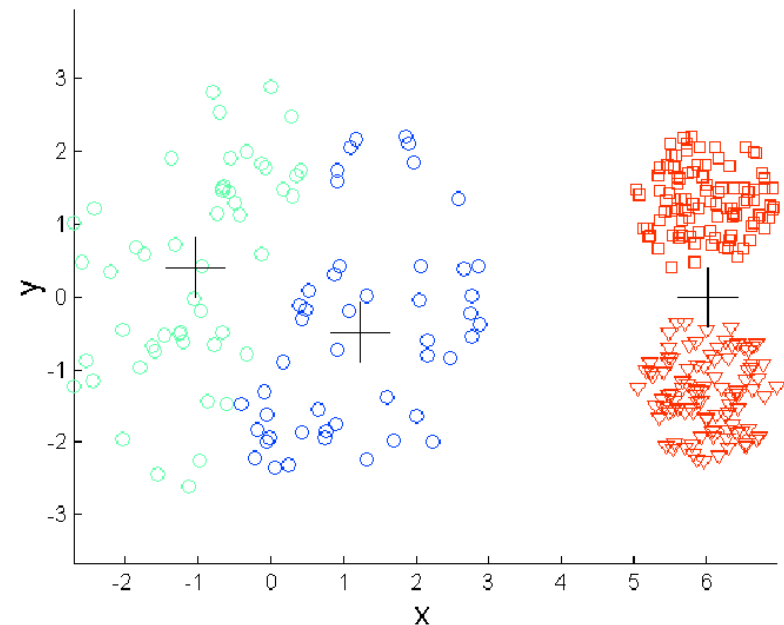
5. Clustering Algorithms

5.1. Partitional Clustering: K-means

- Discussion: Difficult to produce clusters with **different densities**



Original set of points
(colors of points represent desired clusters)



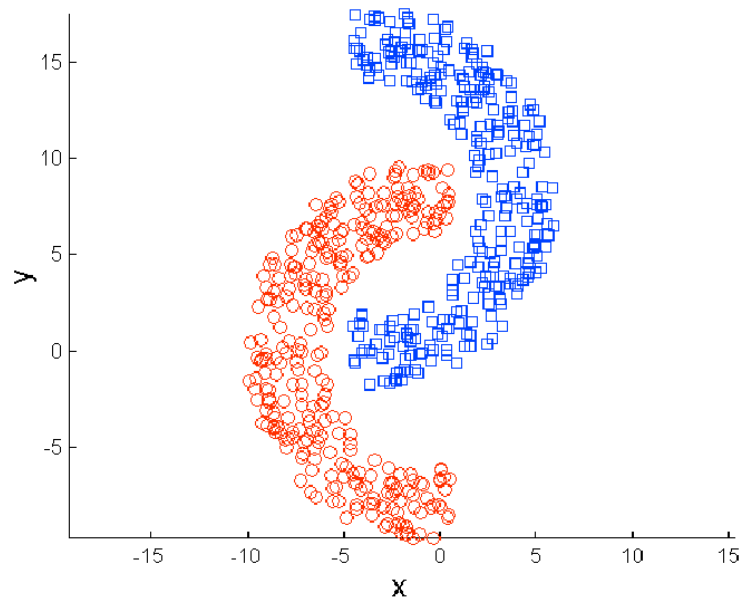
Expected k-means result with k=3

Ch. 10 – Data Clustering

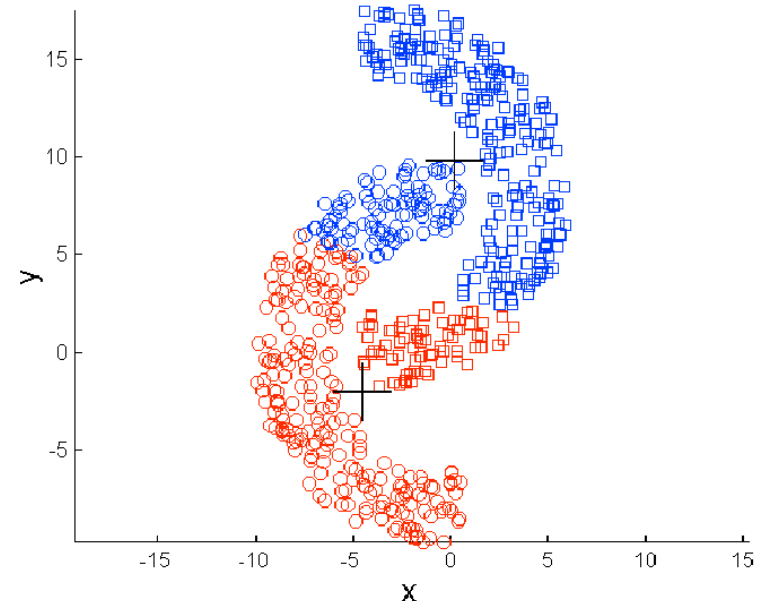
5. Clustering Algorithms

5.1. Partitional Clustering: K-means

- Discussion: Difficult to produce clusters with **non-convex** (non-globular) **shapes**



Original set of points
(colors of points represent desired clusters)



Expected k-means result with k=3

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.1. Partitional Clustering: K-means

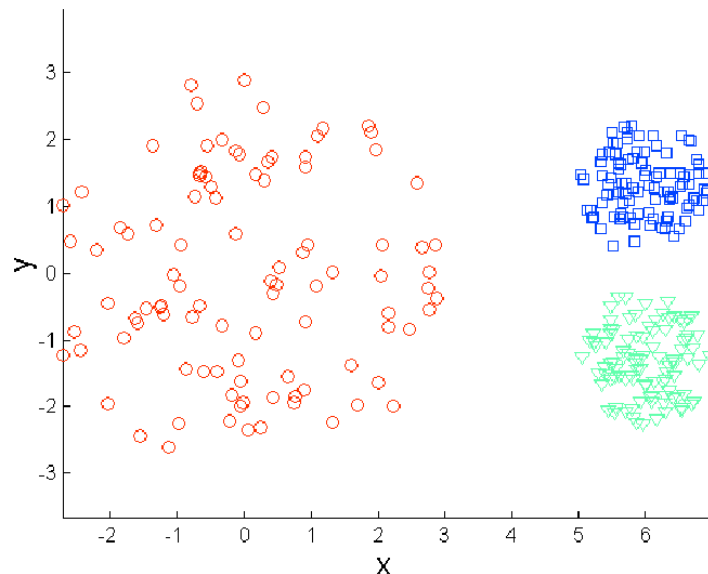
- **Possible solutions** to overcome some of k-means' problems
 1. Execute **multiple runs** of the algorithm
 - Downside: **increase clustering time**
 2. **Select more** than the user desired **k** centroids
 - Then **post-process** result by eliminating small clusters that might represent outliers
 - **Merge** clusters that are close and that have relatively low SSE
 - And maybe **split** “loose” clusters: with relatively high SSE
 3. **Filter out outliers**, i.e., **pre-processing** before running clustering algorithm
 - Using filter (similarity/distance) function
 4. **Informed** (or heuristic) **selection** of initial **k** centroids
 - Instead of random selection

Ch. 10 – Data Clustering

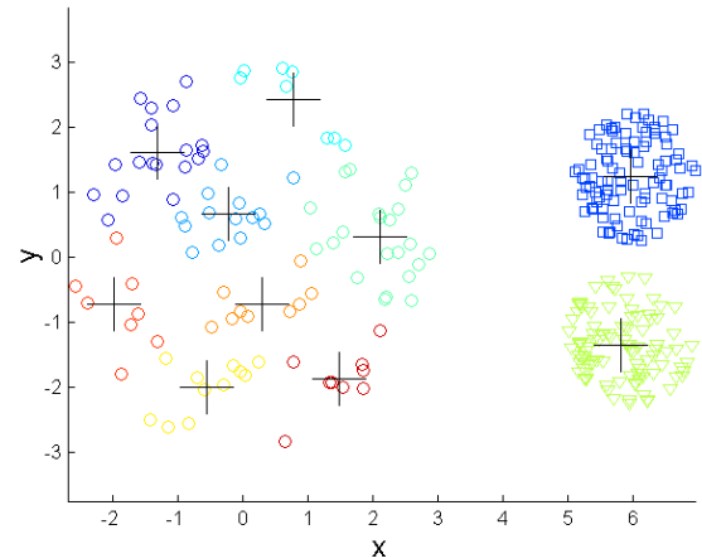
5. Clustering Algorithms

5.1. Partitional Clustering: K-means

- **Possible solutions** to overcome some of k-means' problems



Original set of points
(colors of points represent desired clusters)



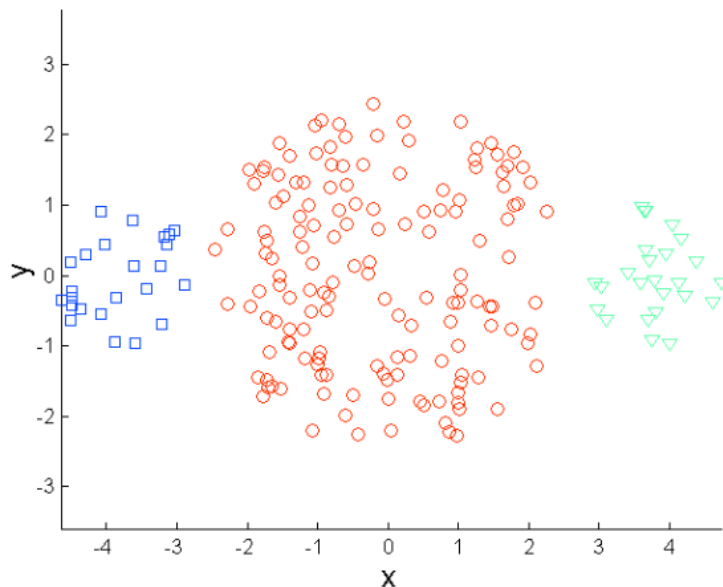
Selecting more than $k=3$ clusters
(producing smaller clusters, which can then be merged into bigger ones: post-processing)

Ch. 10 – Data Clustering

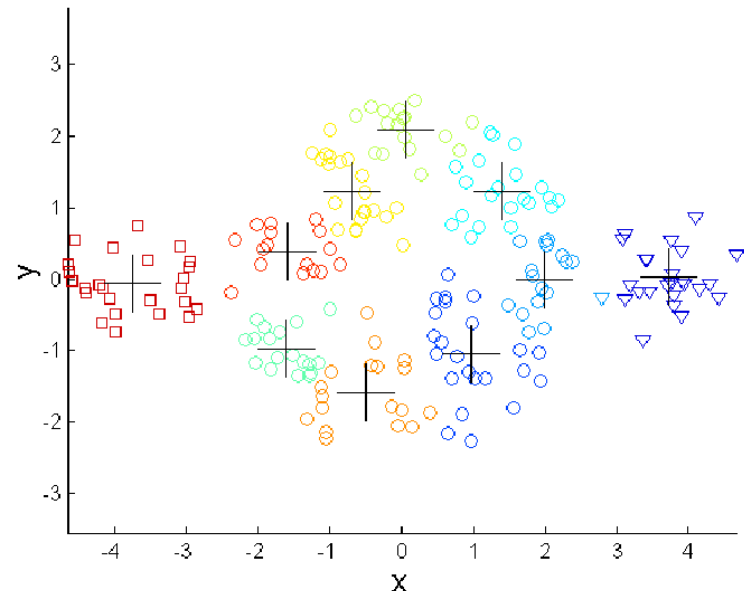
5. Clustering Algorithms

5.1. Partitional Clustering: K-means

- **Possible solutions** to overcome some of k-means' problems



Original set of points
(colors of points represent desired clusters)



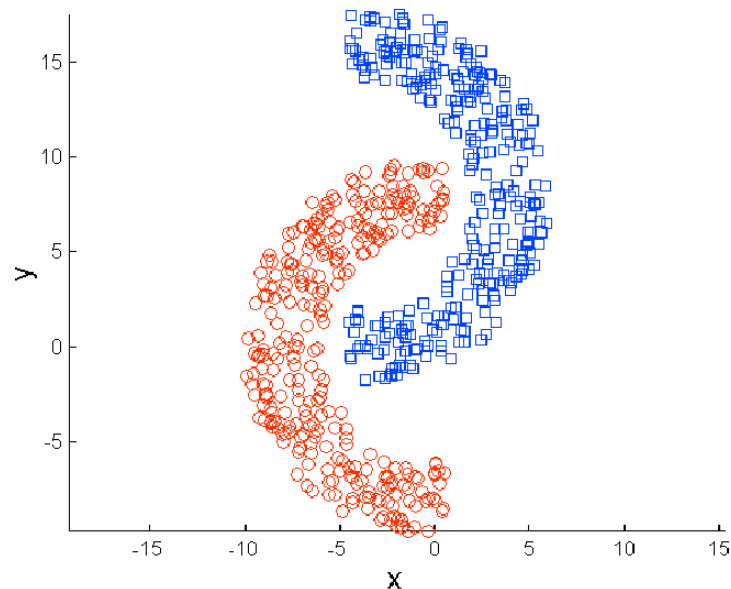
Selecting more than $k=3$ clusters
(producing smaller clusters, which can then be merged into bigger ones: post-processing)

Ch. 10 – Data Clustering

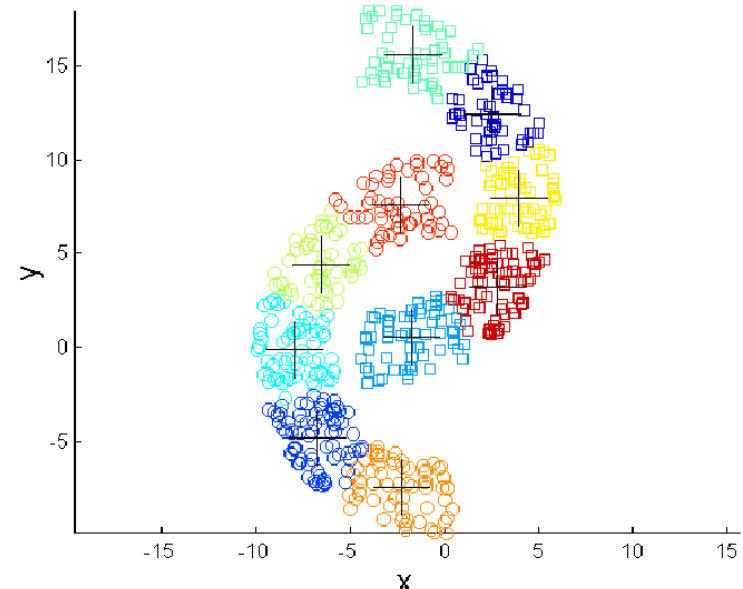
5. Clustering Algorithms

5.1. Partitional Clustering: K-means

- **Possible solutions** to overcome some of k-means' problems



Original set of points
(colors of points represent desired clusters)



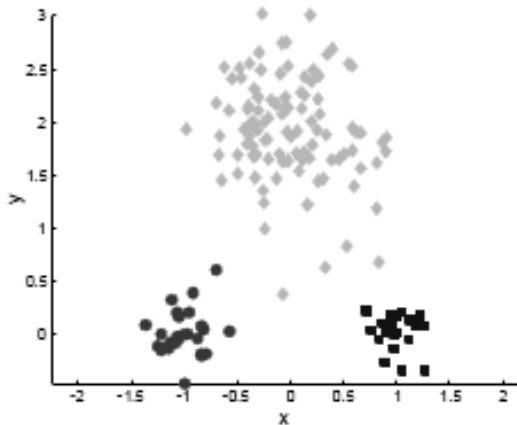
Selecting more than $k=3$ clusters
(producing smaller clusters, which can then be merged into bigger ones: post-processing)

Ch. 10 – Data Clustering

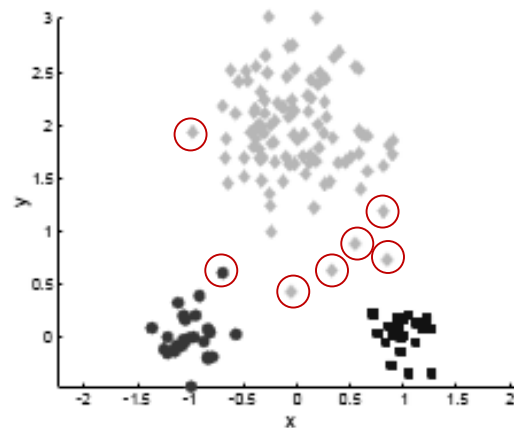
5. Clustering Algorithms

5.1. Partitional Clustering: K-means

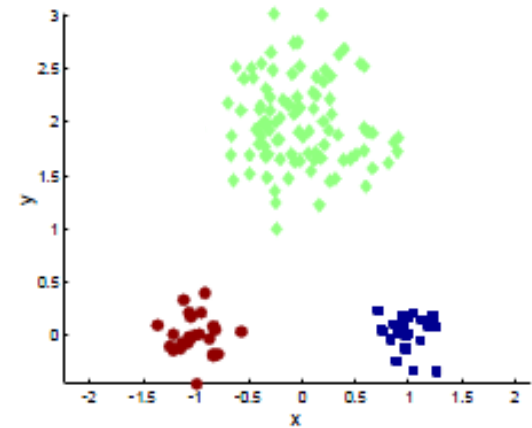
- **Possible solutions** to overcome some of k-means' problems



Original set of points



Filter out outliers
(pre-processing step)



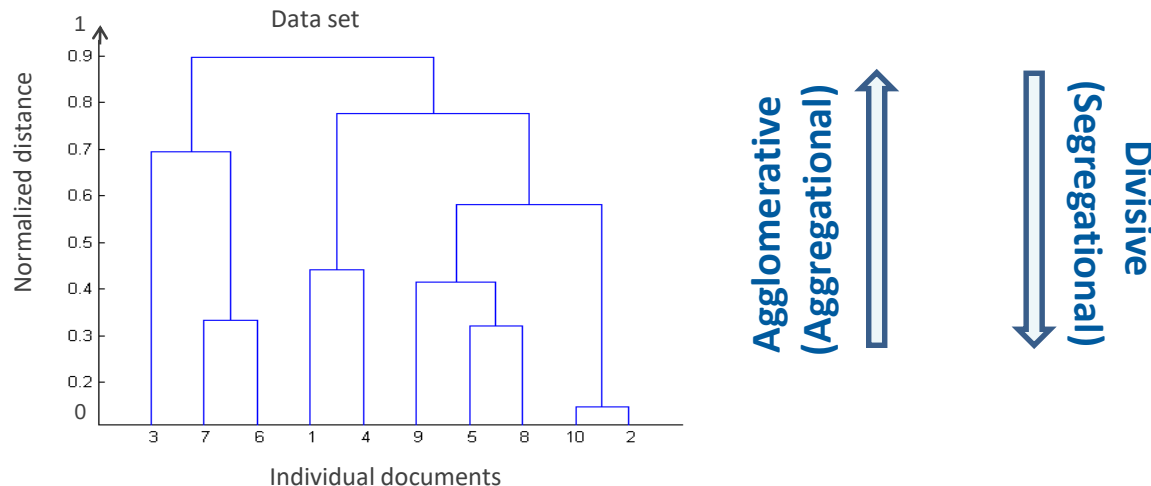
Running k-means with k=3

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering

- **Hierarchical clustering:** builds a hierarchy of clusters, known as a **dendrogram**
 - The **root node** of the dendrogram **represents** the **whole data set** and each **leaf node** **represents** a data **object**
 - ⇒ The dendrogram gives a description and visualization of the clustering process

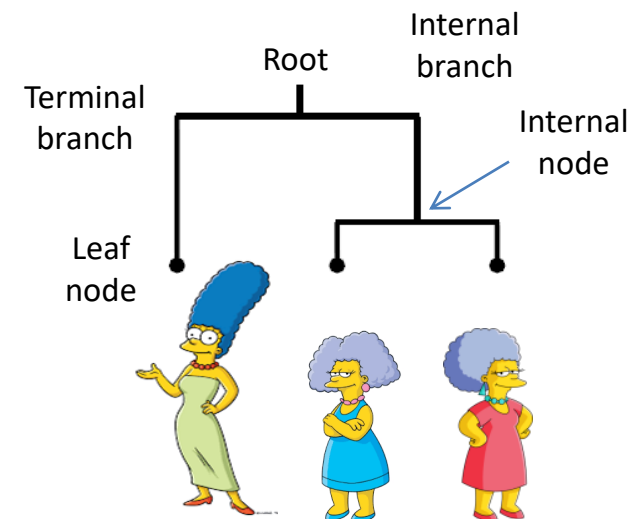


Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering

- To better appreciate and evaluate hierarchical clustering results
 - Let us first introduce the notion of **dendrogram**
- **Dendrogram:** hierarchical tree structure where:
 - **Leaf nodes** represent **data objects**
 - **Internal nodes** represent **clusters of objects**
 - **Similarity** between objects is represented as the **height of their lowest internal node**
 - ⇒ The **more similar**, the **lesser** the **height**

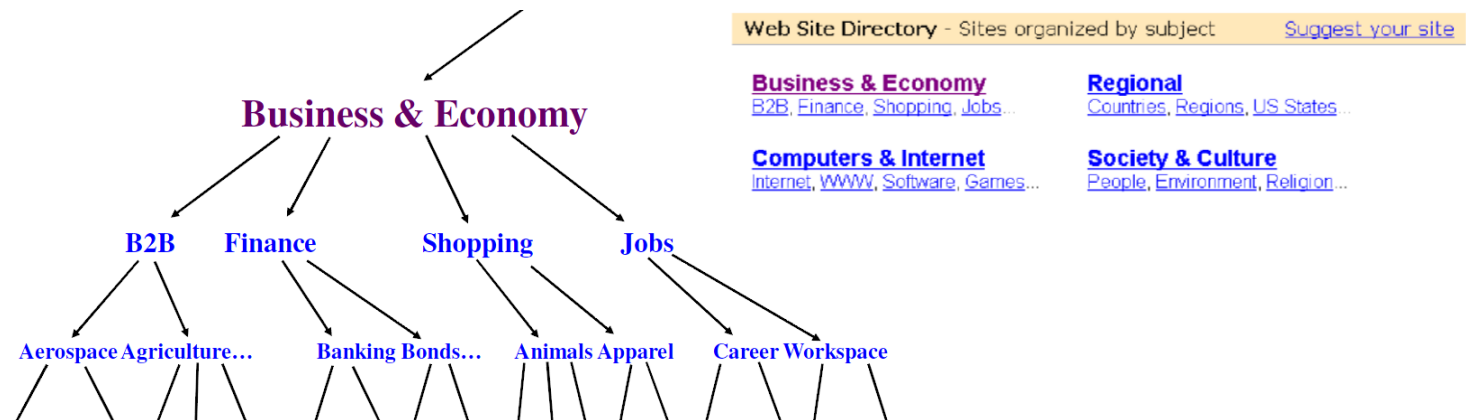


Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering

- Note that hierarchies are commonly used to organize information
 - E.g., Web portals (e.g., Yahoo), semantic taxonomies (e.g., WordNet), etc.
- NOTE: While the latter hierarchies are manually created
 - Dendrograms are automatically created during clustering

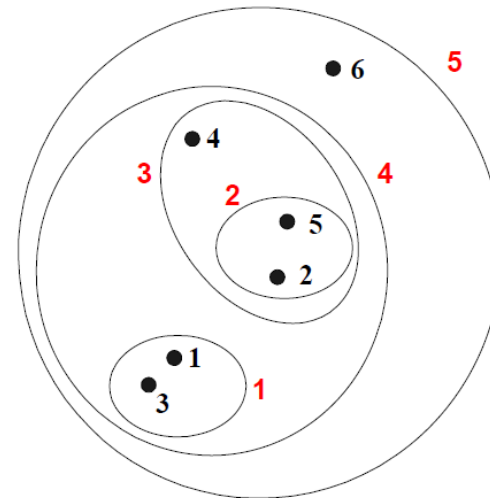
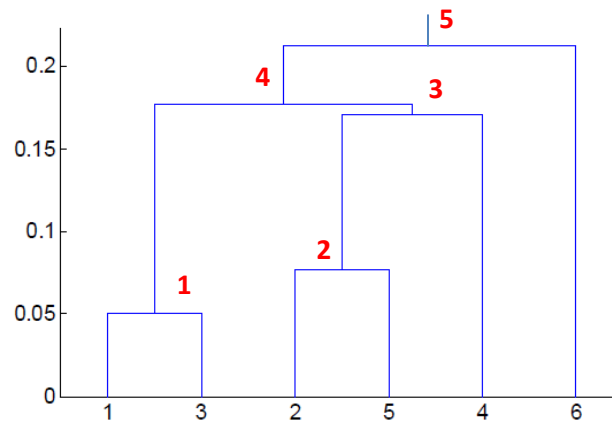


Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering

- Hierarchical clustering consists in producing a set of nested clusters
 - Organized as a hierarchical tree: represented as a **dendrogram**
 - Recording the sequences of **cluster merges** and **splits**



Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering

- We distinguish between two types of hierarchical clustering:
 1. **Agglomerative (Bottom-Up)**: Starting with each object in its own cluster
 - Find the best pair to merge into a new cluster
 - Recursively repeat until all clusters are fused together
 2. **Divisive (Top-Down)**: Starting with all the data objects in a single cluster
 - Consider the best way to divide the cluster into two
 - Recursively repeat on both sides until all clusters are split into objects

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Agglomerative

- **Agglomerative clustering**'s main process:
-

1. Initialization: Compute **inter-cluster similarity matrix**
 2. Let each data object be a cluster
 3. **Repeat**
 4. Grouping: **Merge** the two clusters with maximum similarity
 5. Update: **Re-compute** the **similarity matrix** considering new clusters
 6. **Until** verifying **stopping rule**
-

NOTE: If no stopping rule is defined

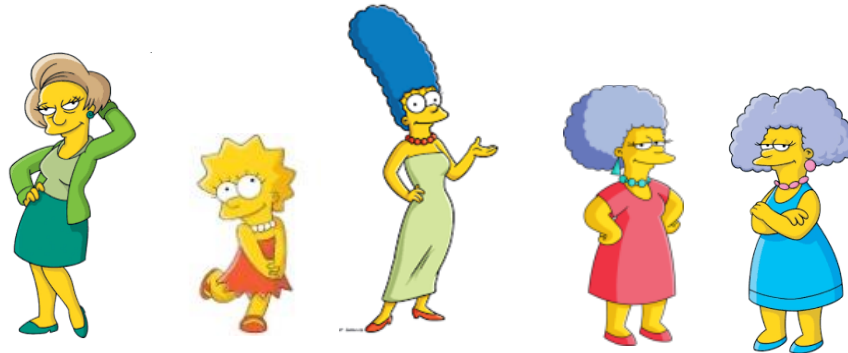
⇒ Algorithm continues until **one single cluster remains**

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Agglomerative

- Example: Consider the following dataset of the Simpson family
 - That we wish to cluster using **agglomerative clustering**



???

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Agglomerative

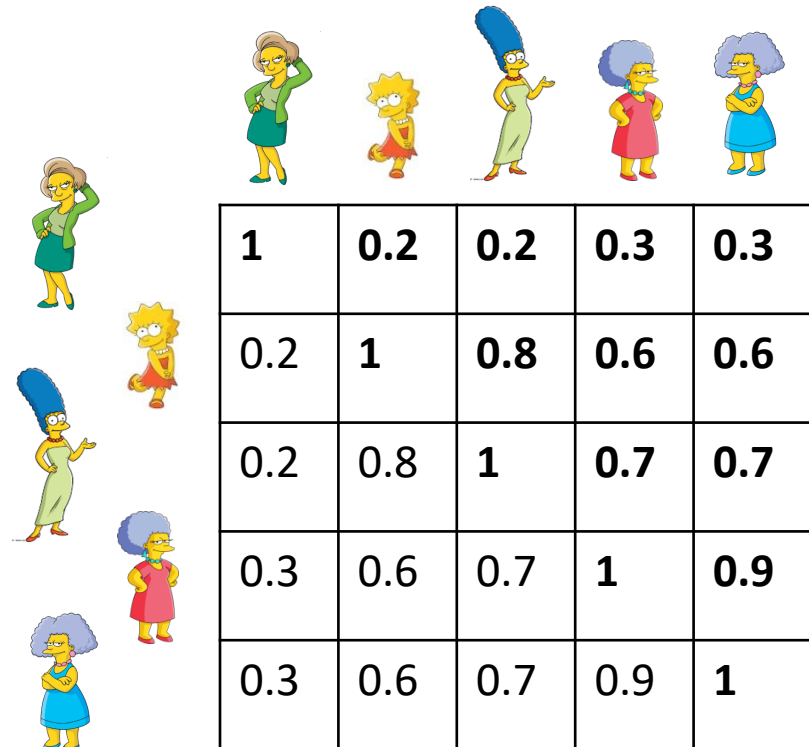
- Example (cont'd): We start by computing the **similarity matrix**
 - Storing the **similarities** (distances) between **every pair of data objects**

- Example:

$$\text{Sim}(\text{Mrs. Simpson}, \text{Mrs. Simpson}) = 1 \text{ (maximum)}$$

$$\text{Sim}(\text{Mrs. Simpson}, \text{Lisa Simpson}) = 0.2 \text{ (low)}$$

$$\text{Sim}(\text{Auntie Plaut}, \text{Auntie Plaut}) = 0.9 \text{ (high)}$$



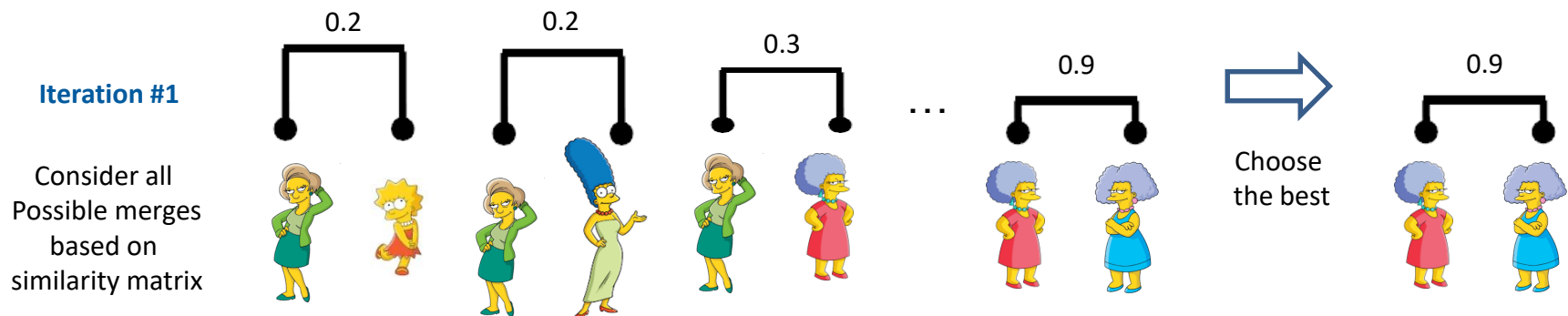
1	0.2	0.2	0.3	0.3
0.2	1	0.8	0.6	0.6
0.2	0.8	1	0.7	0.7
0.3	0.6	0.7	1	0.9
0.3	0.6	0.7	0.9	1

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Agglomerative

- Example (cont'd): We consider all possible merges and execute the best one
 - Merging the closest clusters



Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Agglomerative

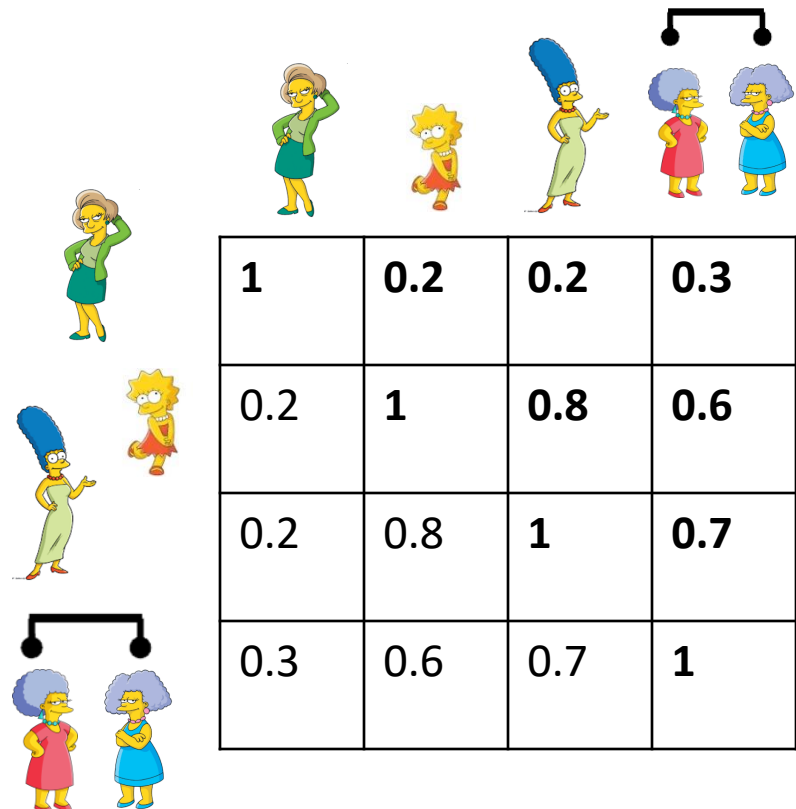
- Example (cont'd): Computing 2nd iteration similarity matrix

– For instance:

$$\text{Sim}(\text{Mrs. Krabappel}, \text{Mrs. Krabappel}) = 1 \text{ (maximum)}$$

$$\text{Sim}(\text{Mrs. Krabappel}, \text{Marge Simpson}, \text{Lisa Simpson}) = 0.3 \text{ (low)}$$

$$\text{Sim}(\text{Marge Simpson}, \text{Marge Simpson}, \text{Lisa Simpson}) = 0.7 \text{ (high)}$$



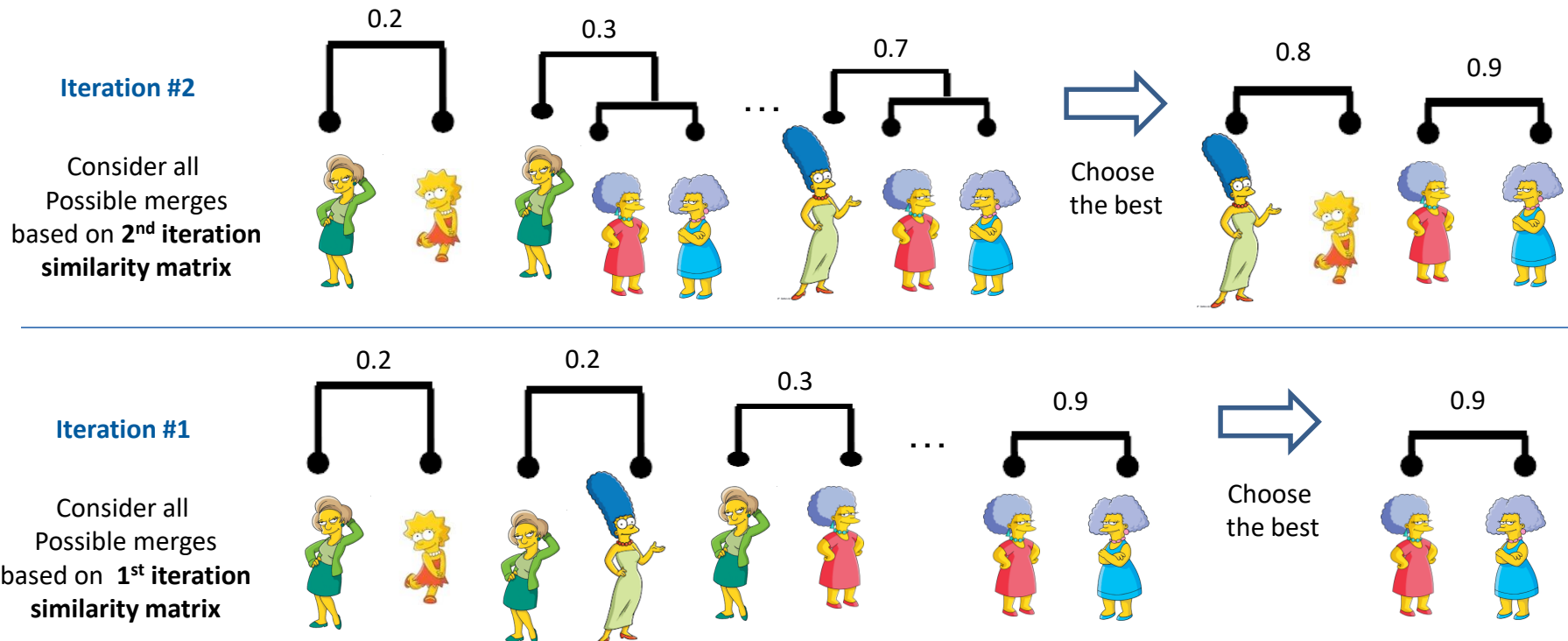
1	0.2	0.2	0.3
0.2	1	0.8	0.6
0.2	0.8	1	0.7
0.3	0.6	0.7	1

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Agglomerative

- Example (cont'd):



Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Agglomerative

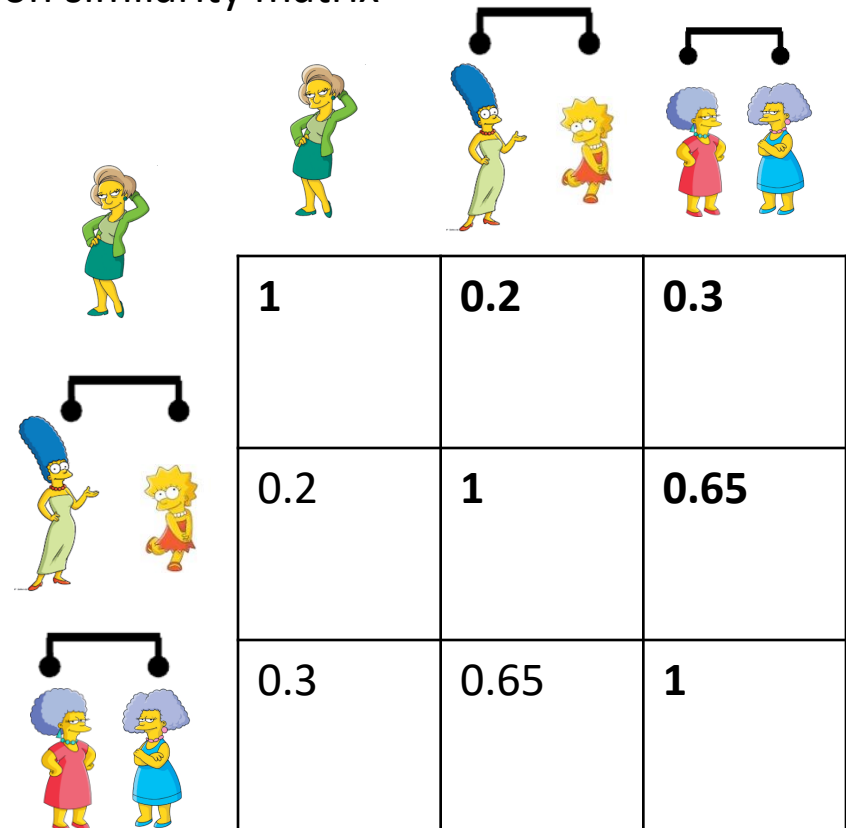
- Example (cont'd): Computing 3rd iteration similarity matrix

– For instance:

$$\text{Sim}(\text{Lisa Simpson}, \text{Lisa Simpson}) = 1 \text{ (maximum)}$$

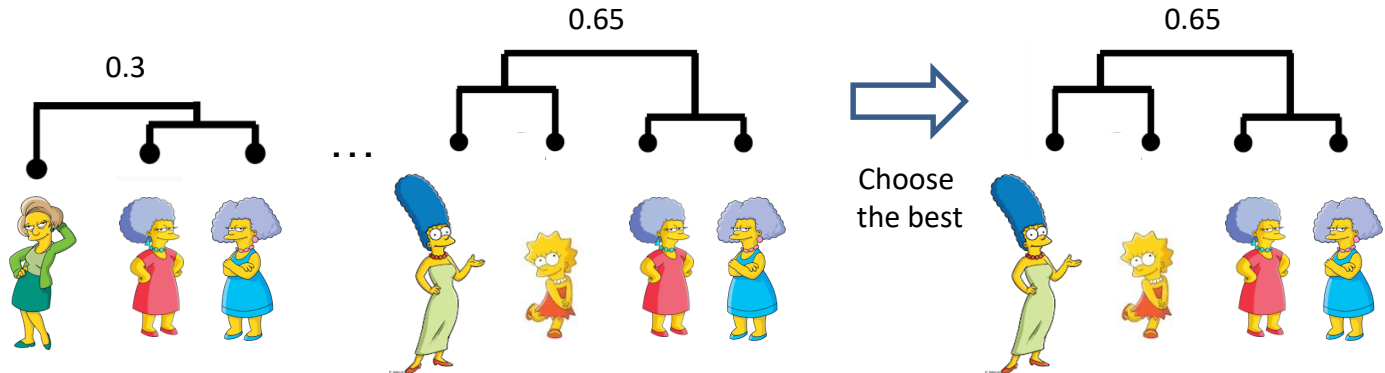
$$\text{Sim}(\text{Lisa Simpson}, \text{Marge Simpson, Bart Simpson}) = 0.3 \text{ (low)}$$

$$\text{Sim}(\text{Marge Simpson, Bart Simpson}, \text{Maggie Simpson, Lisa Simpson}) = 0.65$$



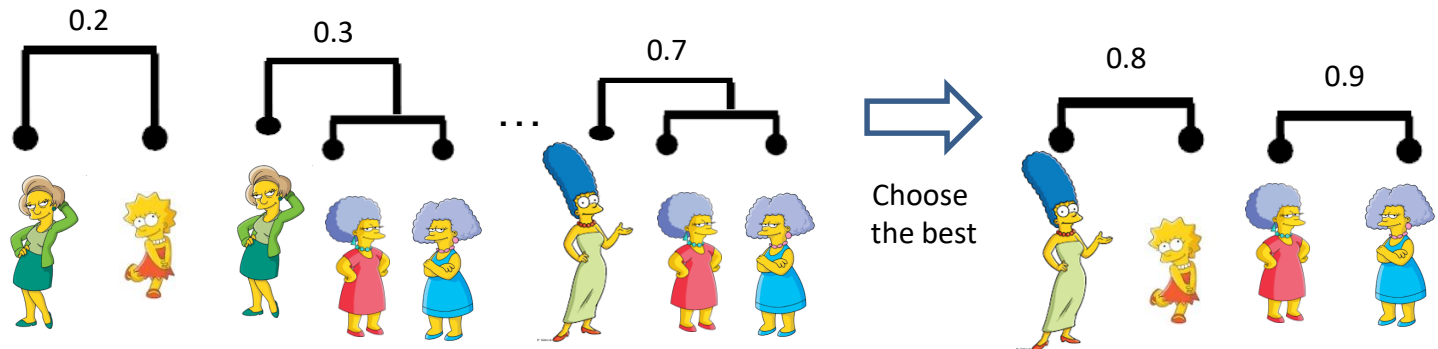
Iteration #3

Consider all
Possible merges
based on **3rd iteration**
similarity matrix



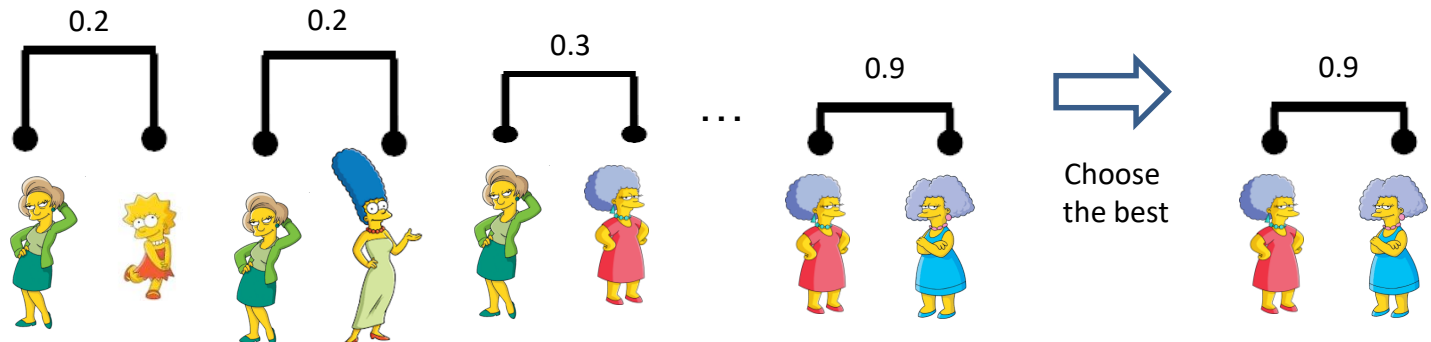
Iteration #2

Consider all
Possible merges
based on **2nd iteration**
similarity matrix



Iteration #1

Consider all
Possible merges
based on **1st iteration**
similarity matrix

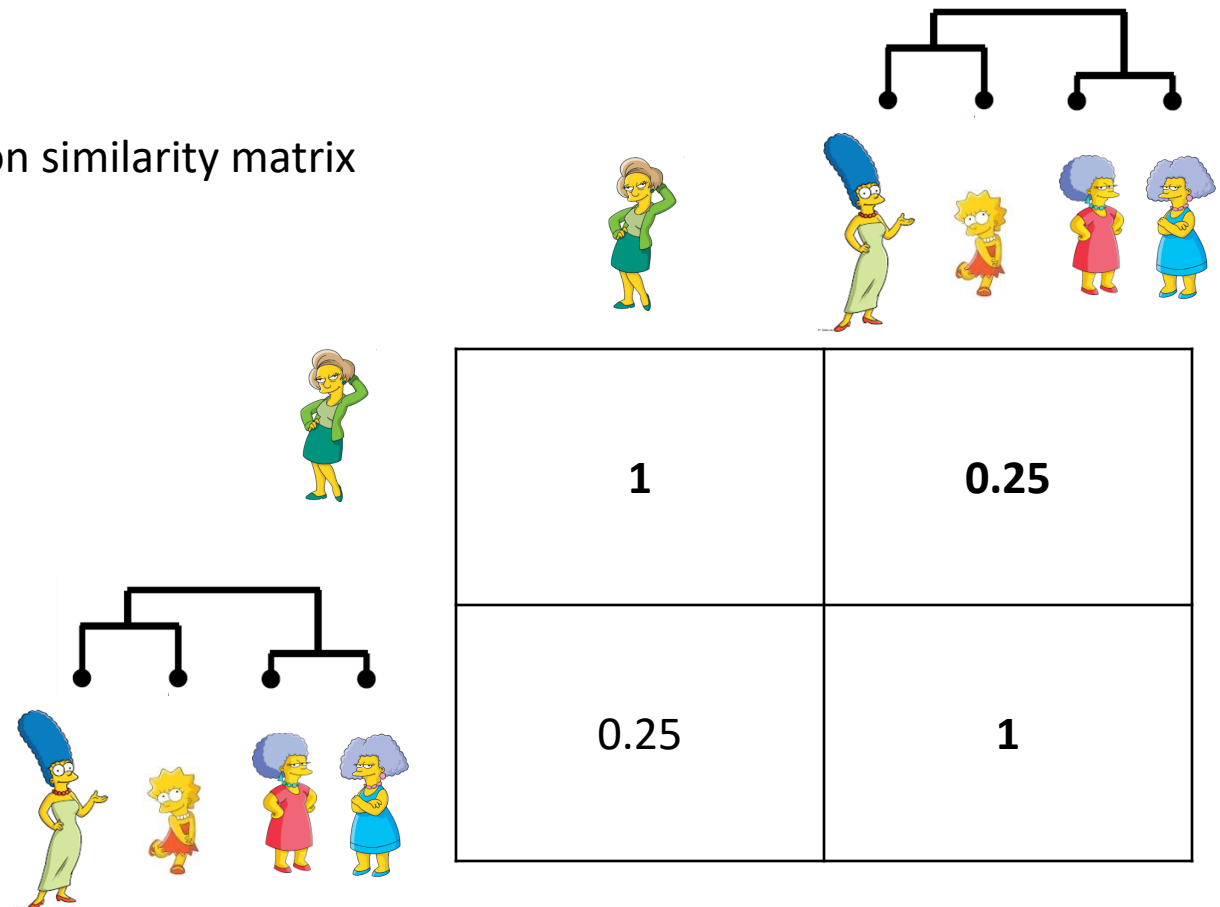


Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Agglomerative

- Example (cont'd):
Computing 4th iteration similarity matrix

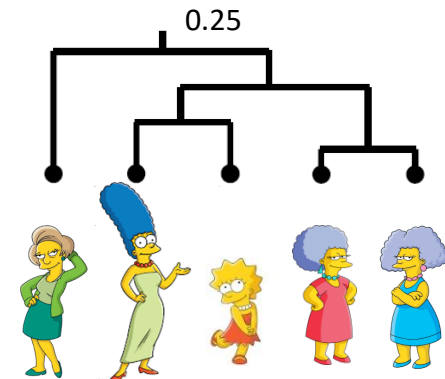


Iteration #4

One possible merge is left between the last two remaining clusters, considering the 4th iteration similarity matrix

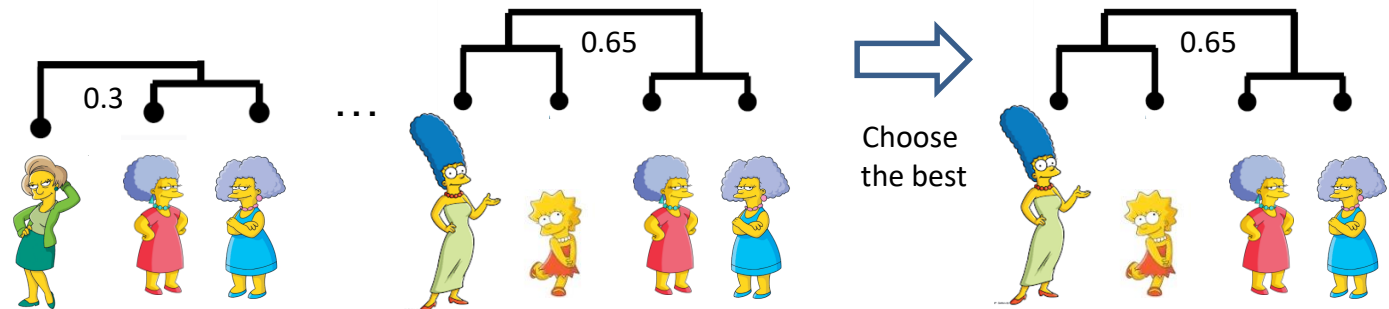
⇒ End of clustering process: **DONE!**

- No stopping rule was defined here
- Hence process stops when merging all objects into one cluster



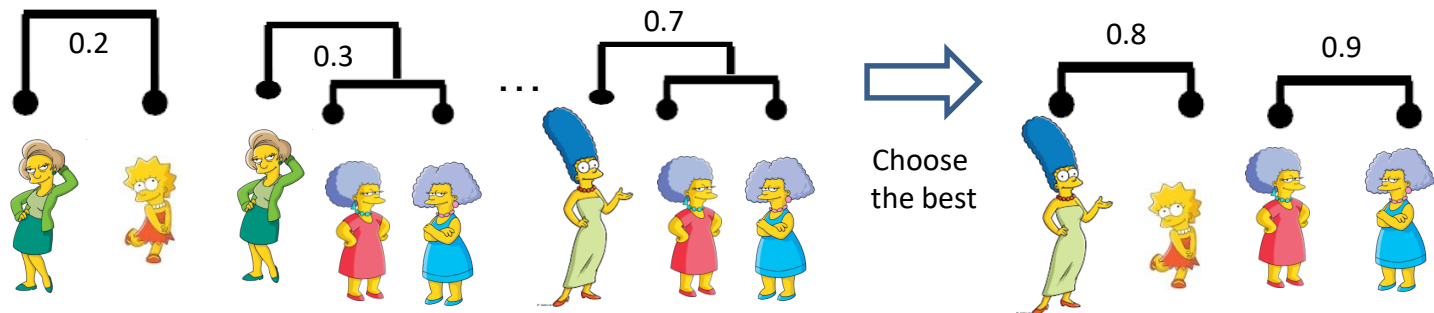
Iteration #3

Consider all Possible merges based on 3rd iteration similarity matrix



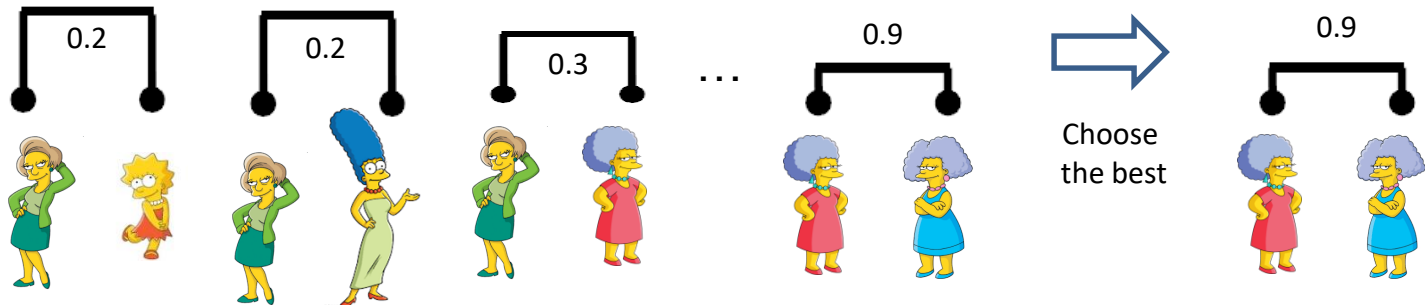
Iteration #2

Consider all Possible merges based on 2nd iteration similarity matrix



Iteration #1

Consider all Possible merges based on 1st iteration similarity matrix



Ch. 10 – Data Clustering

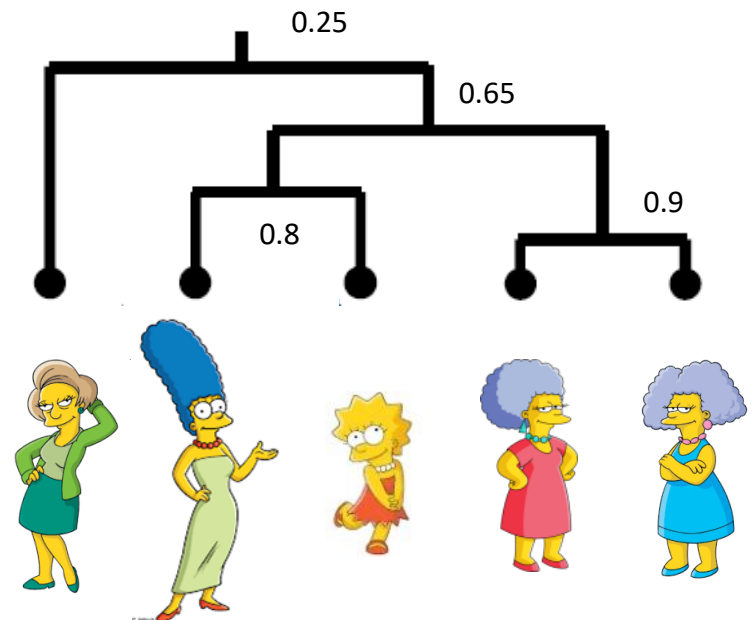
5. Clustering Algorithms

5.2. Hierarchical Clustering: Dendrogram

- Why is the **dendrogram** appealing in hierarchical clustering?

1. It provides a **description** of the clustering process

- Closest objects
- Farthest objects
- How they were grouped



Ch. 10 – Data Clustering

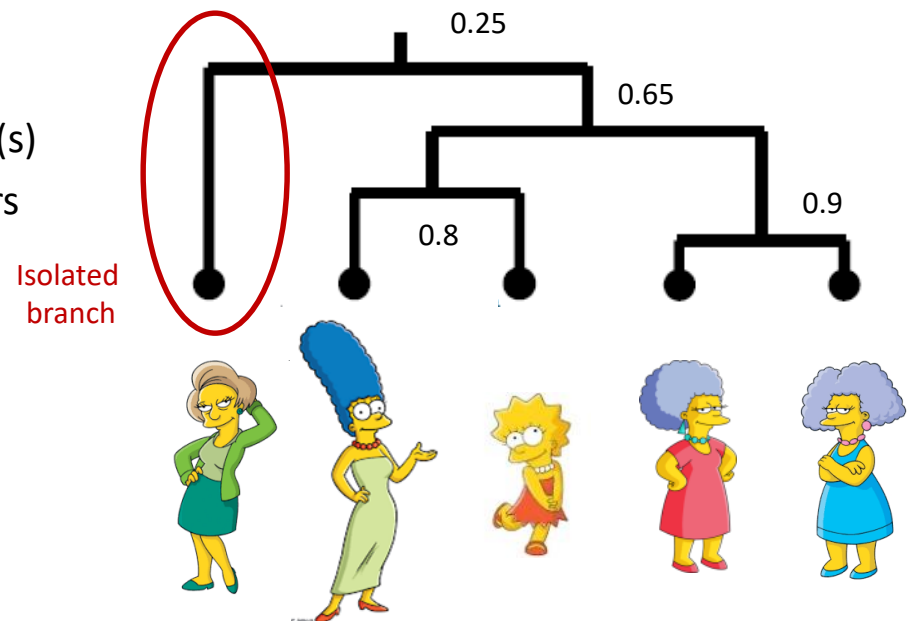
5. Clustering Algorithms

5.2. Hierarchical Clustering: Dendrogram

- Why is the **dendrogram** appealing in hierarchical clustering?

2. It can be used to **detect outliers**

- The most isolated branch(es) is (are) suggestive of (a) data point(s) that is (are) different from all others



Ch. 10 – Data Clustering

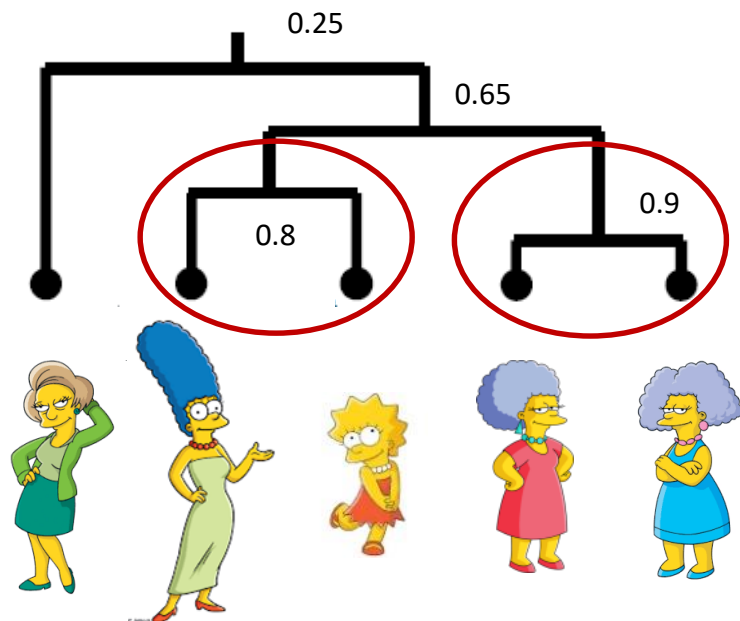
5. Clustering Algorithms

5.2. Hierarchical Clustering: Dendrogram

- Why is the **dendrogram** appealing in hierarchical clustering?

3. It can be used to define a **stopping rule** in order to determine the best set of output clusters

- Highly separated and low-height sub-trees are highly suggestive of good clusters
- High separation** reflects low **inter-cluster similarity**
- Low height** reflects high **intra-cluster similarity**

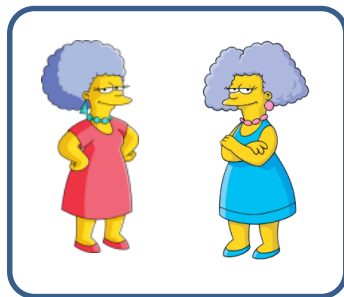


Ch. 10 – Data Clustering

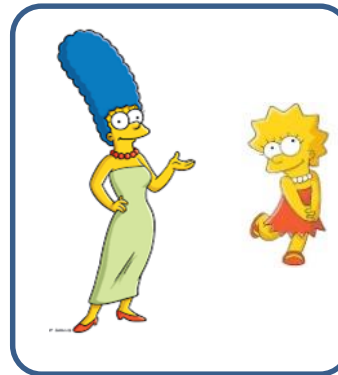
5. Clustering Algorithms

5.2. Hierarchical Clustering: Stopping Rule

- Example (cont'd): Final result of running agglomerative clustering
 - **Ordered** (optional) following **intra-cluster** and **inter-cluster** similarity
 - The clusters with **highest intra-cluster similarity** (most coherent)
 - and the **lowest average inter-cluster similarity** with all other clusters (most distinct)
 - ⇒ comes first



Cluster 1



Cluster 2

Cluster 3
(outlier)

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Intra & Inter Cluster Similarity

Different formulas exist to evaluate intra and inter-cluster similarity, e.g.:

- **Intra-cluster** similarity (1)

- Usually evaluated as **Sum of Squared Error** (or SSE, cf. *k-means*)

$$\Rightarrow SSE = \sum_{i=1}^k \sum_{x_p \in C_i} Dist(x_p, m_i)^2 \quad \text{where } Dist \text{ is the inverse function of } Sim$$

- k is the number of clusters at the designate hierarchical level (iteration)
- m_i is the centroid of cluster C_i
 - Computed as the mean of data objects in C_i

$$\Rightarrow m_i = \frac{1}{|C_i|} \sum_{x_p \in C_i} x_p$$

$\Rightarrow$ **Lower SSE** means **higher intra-cluster similarity!**

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Intra & Inter Cluster Similarity

Different formulas exist to evaluate intra and inter-cluster similarity, e.g.:

- **Intra-cluster** similarity (2)
 - Another measure is **Pair Group Method Average** (or PGMA):

$$\Rightarrow \text{PGMA} = \sum_{i=1}^k \frac{\sum_{x_p \in C_i} \sum_{x_q \in C_i, p \neq q} \text{Sim}(x_p, x_q)}{N_i}$$

- Where k is the number of clusters at the designate hierarchical level (iteration)
- N_i is the number of pair-wise comparison operations within cluster C_i

$$\Rightarrow N_i = \frac{|C_i| \times (|C_i| - 1)}{2}$$

$\Rightarrow$ **Higher PGAM** means **higher intra-cluster similarity!**

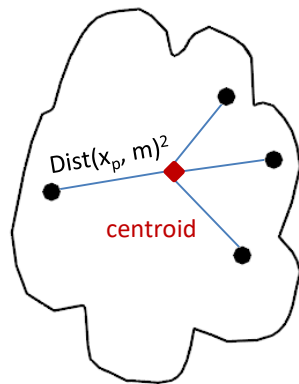
Ch. 10 – Data Clustering

5. Clustering Algorithms

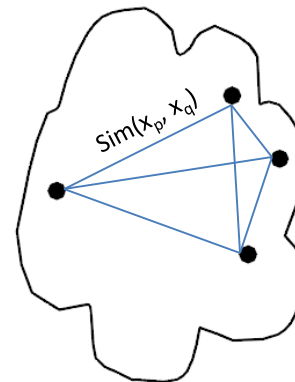
5.2. Hierarchical Clustering: Intra & Inter Cluster Similarity

Different formulas exist to evaluate intra and inter-cluster similarity, e.g.:

- **Intra-cluster** similarity:



SSE



PGMA

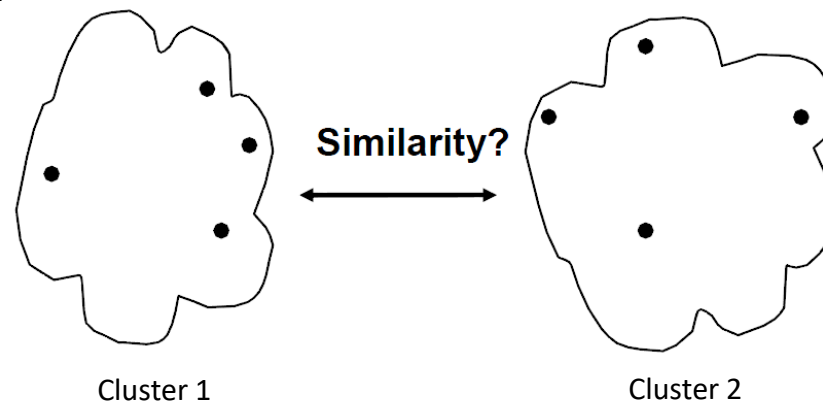
Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Intra & Inter Cluster Similarity

Different formulas exist to evaluate intra and inter-cluster similarity, e.g.:

- Similarly, **inter-cluster** similarity
 - Can be evaluated in different ways:
 - Single Link (Max)
 - Complete Link (Min)
 - Average Link (Avg)



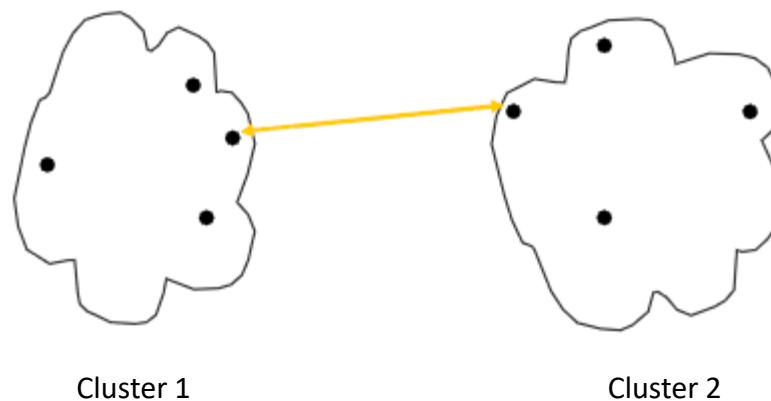
Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Intra & Inter Cluster Similarity

Different formulas exist to evaluate intra and inter-cluster similarity, e.g.:

- **Inter-cluster** similarity (1)
 - Using **Single Link** method
 - Cluster similarity = similarity of **two most similar** cluster objects



(+)
Can handle
Non-globular
shapes

(-)
Produces
potentially
long and skinny
clusters

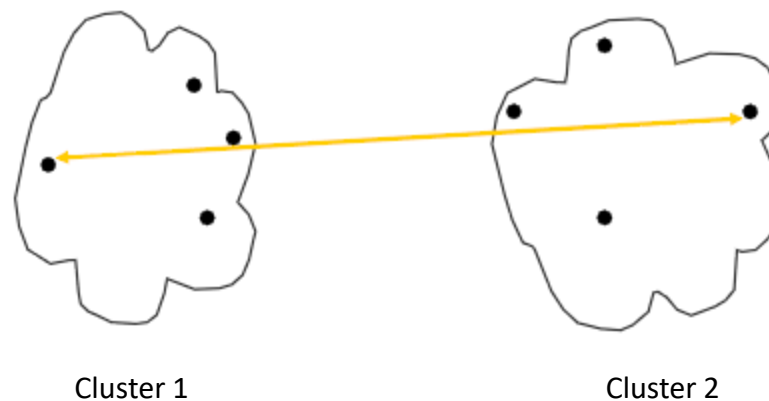
Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Intra & Inter Cluster Similarity

Different formulas exist to evaluate intra and inter-cluster similarity, e.g.:

- **Inter-cluster** similarity (2)
 - Using **Complete Link** method
 - Cluster similarity = similarity of **two least similar** cluster objects



(+)
Produces
potentially
compact
clusters

(-)
Tends to
break large
clusters

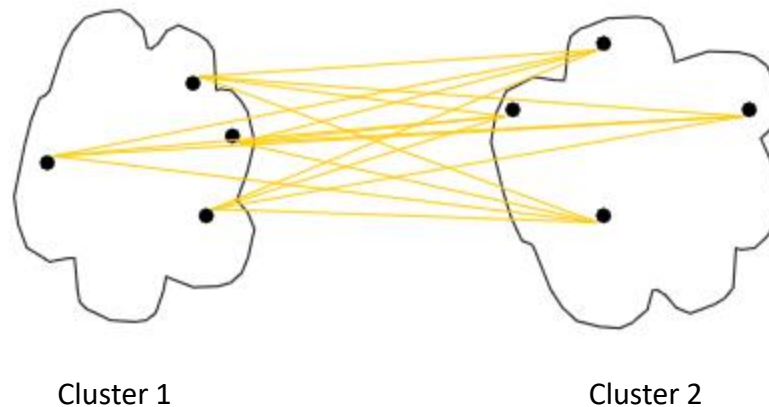
Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Intra & Inter Cluster Similarity

Different formulas exist to evaluate intra and inter-cluster similarity, e.g.:

- **Inter-cluster** similarity (3)
 - Using **Average Link** method
 - Cluster similarity = **average similarity** of **all pairs** of objects from each cluster



(+)
Most **robust**
against noise,
and most
widely used

(-)
Biased toward
globular clusters

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Divisive

- So far we have only looked at **agglomerative (bottom-up) clustering**
 - But a cluster hierarchy can also be generated **top-down**
 - Using so-called **divisive clustering**
 - We start at the top with all documents in **one cluster**
 - The cluster is **split** using a **flat clustering** algorithm (e.g., k-means)
 - This procedure is applied recursively until each document is in its own singleton cluster
- ⇒ **Top-down** clustering is conceptually **more complex** than **bottom-up**!
- Since we need a second, flat clustering algorithm as a ``subroutine''

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.2. Hierarchical Clustering: Divisive

- **Divisive clustering**'s main process:
-

1. Initialization: Compute **intra-cluster similarity matrix**
 2. Let all data object in dataset be one single cluster
 3. **Repeat**
 4. Dividing: **Bisect** the cluster with lowest intra-cluster similarity
 5. Update: **Re-compute** the **similarity matrix** considering new clusters
 6. **Until** verifying **stopping rule**
-

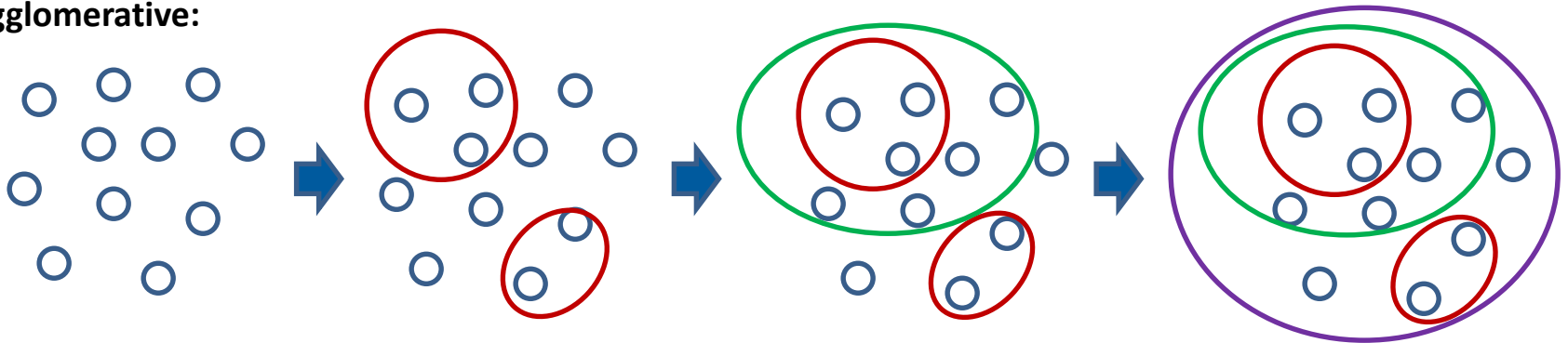
- NOTE:**
1. Bisection is done using a **flat clustering** sub-routine (e.g., *k-means* with $k = 2$)
 2. If no stopping rule is defined, algorithm continues until
 - ⇒ **each data object** forms an **individual cluster**

Ch. 10 – Data Clustering

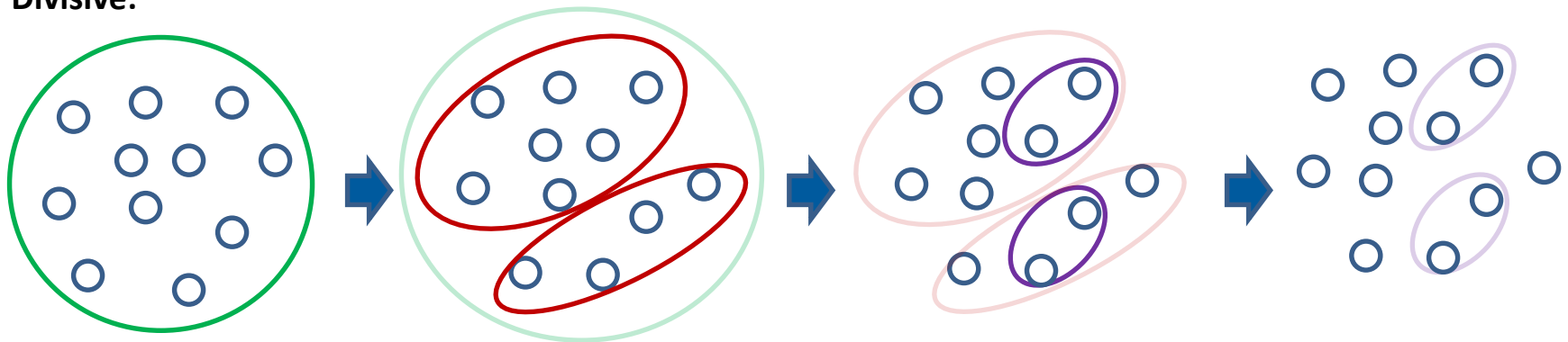
5. Clustering Algorithms

5.2. Hierarchical Clustering: Divisive

Agglomerative:



Divisive:



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5. Clustering Algorithms

5.2. Hierarchical Clustering

- **Pros** of hierarchical clustering:
 - **No need** to specify **number of clusters** in advance
 - Hierarchical structure (**dendrogram**) maps nicely onto **human intuition**
- **Cons** of hierarchical clustering:
 - Does **not scale well**: of at least $O(n^2 \log(n))$ time and $O(n^2)$ space complexity, where **n** is # of objects
 - Once a **decision** is made **to combine** two clusters: it **cannot be undone**
 - Different schemes have different issues:
 - Can be biased toward **non-globular clusters**
 - Can be sensitive to **noise**
 - Might tend to **break large clusters**

Ch. 10 – Data Clustering

5. Clustering Algorithms

Comparing **hierarchical** with **partitional** clustering:

Hierarchical algorithms	Partitional algorithms
More complex: $O(n^2 \times \log(n))$ time, and $O(n^2)$ space	Less complex: $O(k \times n \times i)$ time, and $O(n)$ space
Do not usually require any input	Usually require input: number of initial clusters or initial centroids
Provides description of clustering process in the form of a dendrogram	No description of clustering process
Generally produce higher quality clusters	Lower quality clusters in comparison with hierarchical methods

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5. Clustering Algorithms

5.3. *Other Approaches*

- **Incremental** clustering
- **Constrained agglomerative** clustering
- **Density-based** clustering
- **Spectral** clustering
- **Fuzzy** clustering

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.3. Other Approaches: Incremental Clustering

- **Incremental clustering:** Considering a stream of data objects (e.g., XML documents, fragments, or elements)
 - Each data object has to be processed one at a time
 - And assigned to existing clusters
- Overall process:

 1. The **first data** object is assigned to an **original cluster**
 2. The following object is either:
 - 2.1. Assigned to one of the existing clusters
 - Based on the highest **similarity between** the **object** and **existing clusters**
 - 2.2. Assigned to a new cluster
 - If similarity with all clusters is **below** a certain **threshold**
 3. **Repeat step 2** until all data objects are assigned to specific clusters

5. Clustering Algorithms

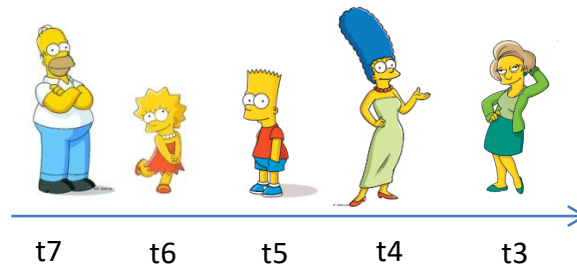
5.3. Other Approaches: Incremental Clustering

- Example:

And so on...

Iteration #3

3rd incoming object is assigned to most similar cluster

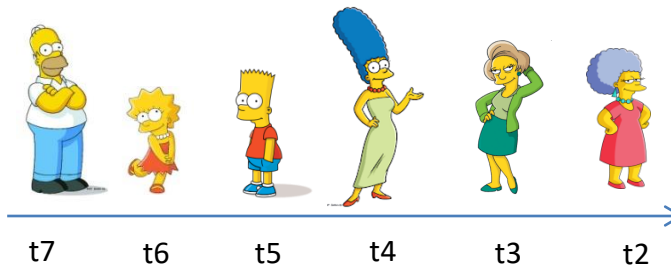


Similarity below threshold, so object forms new cluster



Iteration #2

2nd incoming object is assigned to most similar cluster

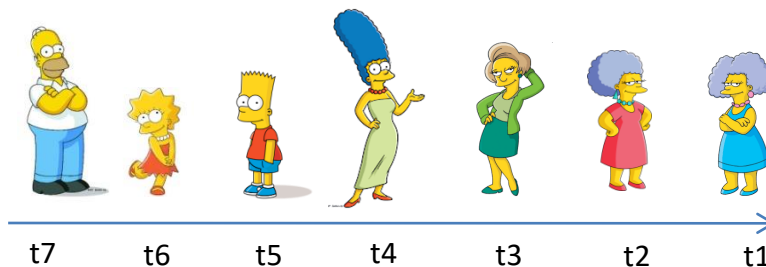


Clustering object in most similar cluster



Iteration #1

1st incoming object is assigned to original cluster



Original Cluster Formed at t1



Stream of incoming objects

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5. Clustering Algorithms

5.3. Other Approaches: Incremental Clustering

- Incremental clustering algorithms are **non-iterative**
 - **Pros:**
 - **Intuitive** and easy to implement
 - They do not require to compute the similarity between each pair of objects
⇒ **Faster** than typical approaches, requires average $O(n)$ time
 - **Cons:**
 - Yet they usually produce a **lower quality** solution
 - Depending on the **order** following which data objects are being received/clustered
 - They share mostly the **same limitations** of **partitional** approaches

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5. Clustering Algorithms

5.3. Other Approaches: Constrained Agglomerative

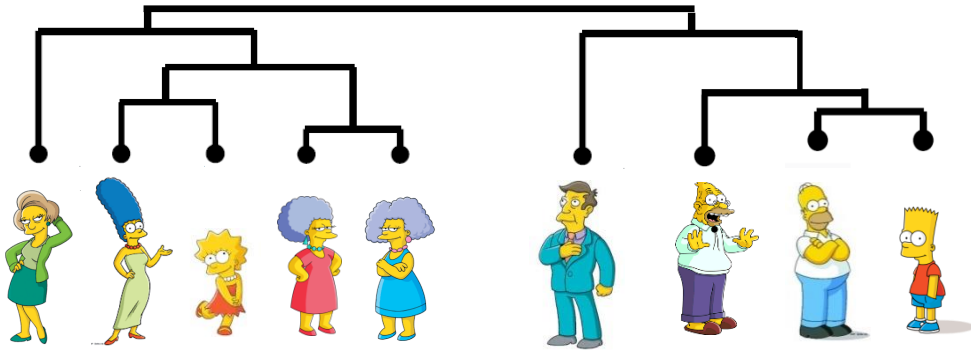
- **Constrained agglomerative:** trade-off between hierarchical and non-hierarchical
- Overall process:

 1. Using **partitional clustering** to constrain the initial objects data set
 2. Each of the produced clusters is treated as a separate set
 - Referred to as **constrained clusters**
 - Objects are only allowed to merge with others that are part of the same set
 1. An **agglomerative algorithm** is used to build a dendrogram tree for each set
 2. Finally, the different dendrogram trees can be combined into a single tree

- Example:

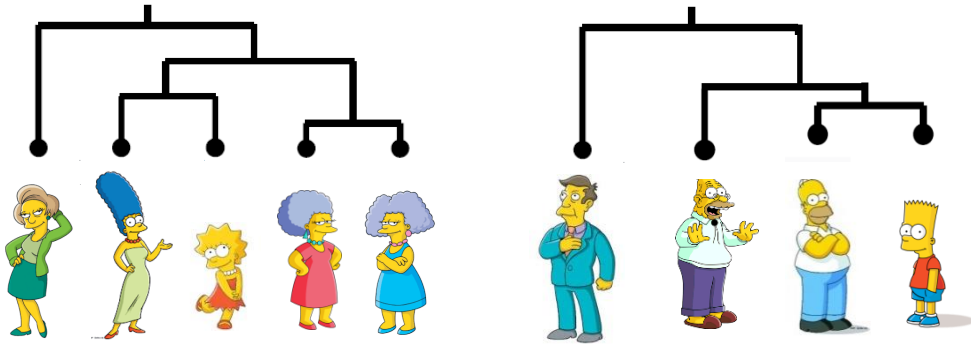
Step 3

Merging dendrograms



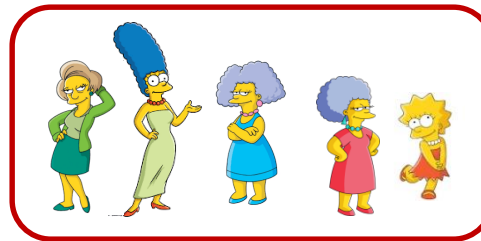
Step 2

An agglomerative clustering is used to produce dendrograms for every constrained set

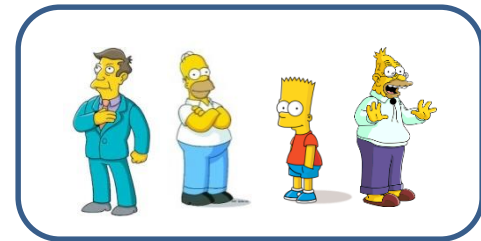


Step 1

Using partitionial clustering to constrain initial data set

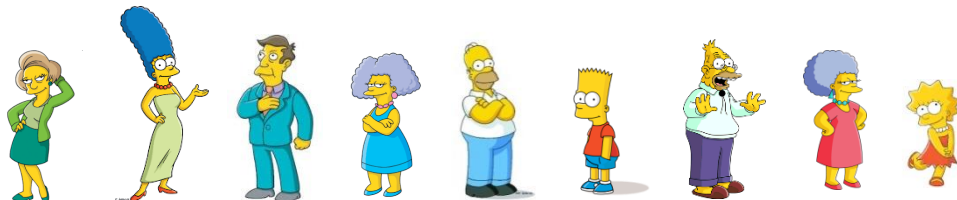


Cluster 1



Cluster 2

Original data set



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5. Clustering Algorithms

5.3. Other Approaches: Constrained Agglomerative

- **Constrained agglomerative:** trade-off between hierarchical and
- **Pros:**
 - Considering user(environment)-specific constraints to start with
 - Computational complexity can be reduced to $O(n^2)$
- **Cons:**
 - Depends on input constraints (which might not be available)
 - Cluster quality depends on the choice of user constraints

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.3. Other Approaches: Spectral Clustering

- **Spectral clustering:** using the spectrum (eigen values) of the similarity matrix of the data set to perform dimension reduction
 - Before clustering in reduced (and maybe more descriptive) dimensions
- Overall process

 1. Compute **similarity matrix** S between each pair of data objects
 2. Performing **Latent Semantic processing** (e.g., **singular value decomposition** or **principal component analysis**)
 - $S = U \Sigma V^T$ where Σ designates reduced feature space dimensions
 3. Clustering is performed in the **reduced feature space**
 - Using an existing clustering approach

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5. Clustering Algorithms

5.3. Other Approaches: Spectral Clustering

- Main concept of dimension transformation/reduction techniques:
 - Such as *singular value decomposition* (SVD), principal component analysis (PCA), or *initial component analysis* (ICA)
 - Transform **object/object** similarity matrix into **object/concept** matrix
 - where concepts represent new (latent semantic) dimensions

$$\begin{array}{ccccccc}
 & o & & c & & c & & o \\
 o \begin{bmatrix} & \end{bmatrix} & = & o \begin{bmatrix} & \end{bmatrix} & \times & c \begin{bmatrix} & \end{bmatrix} & \times & c \begin{bmatrix} & \end{bmatrix} \\
 \text{Object/object} & & \text{Object/concept} & & \text{Co-variance} & & \text{Concept/object} \\
 \text{similarity matrix} & & \text{matrix} & & \text{matrix} & & \text{matrix}
 \end{array}$$

- ⇒ Then clustering is performed on **object/concept** matrix
- Representing **transformed** (semantic) object **space**

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5. Clustering Algorithms

5.3. Other Approaches: Spectral Clustering

- **Spectral clustering** will share some of the same **pros** and **cons**
 - of the algorithm adopted to perform the clustering in the reduced feature space

In addition:

- **Pros:** - Transformed feature space can be more descriptive of data objects
 - ⇒ Producing **higher quality** results
 - Reduced feature space
 - ⇒ Allowing more efficient similarity computations
- **Cons:** - Semantic feature space produced with **latent semantic processing** is not related to user's perception of semantics
 - Implicit semantic dimensions
 - ⇒ Result **might not meet the user's needs**
 - Difficult for the user to decide on the parameters and dimensionality of the latent semantic representation

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5. Clustering Algorithms

5.3. Other Approaches: Density-based Clustering

- **Density-based clustering** groups objects that **are closely packed** together
 - I.e., objects with many **nearby neighbors**
 - And marks as **outliers** objects that lie alone in low-density regions
- **DBSCAN** is the main density-based algorithm used in the literature
 - It accepts as input:
 1. A neighborhood radius: ϵ
 2. A minimum number of points necessary to form a cluster: ***minPts***

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5. Clustering Algorithms

5.3. Other Approaches: Density-based Clustering

- Using ϵ and **minPts**, **DBSCAN** categorizes objects into:
 1. **Core Points:** A data point p is a *core point* if its ϵ -neighborhood $N_\epsilon(p)$ contains at least *minPts*: $|N_\epsilon(p)| \geq \text{minPts}$, including the point p itself
 2. **Border Points:** A data point q is a *border point* if $N_\epsilon(q)$ contains less than *minPts* data points, but q is **reachable** from some *core point* p
 - A point q is reachable from p if there is a path $p_1, \dots, p_n$ with $p_1 = p$ and $p_n = q$, where each p_{i+1} is **directly reachable** from p_i
 - A point p is **directly reachable** from q if it is in q 's neighborhood (or vice versa)
 3. **Outliers:** A data point o is an *outlier* if it is neither a core point nor a border point

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5. Clustering Algorithms

5.3. Other Approaches: Density-based Clustering

- DBSCAN's main process:

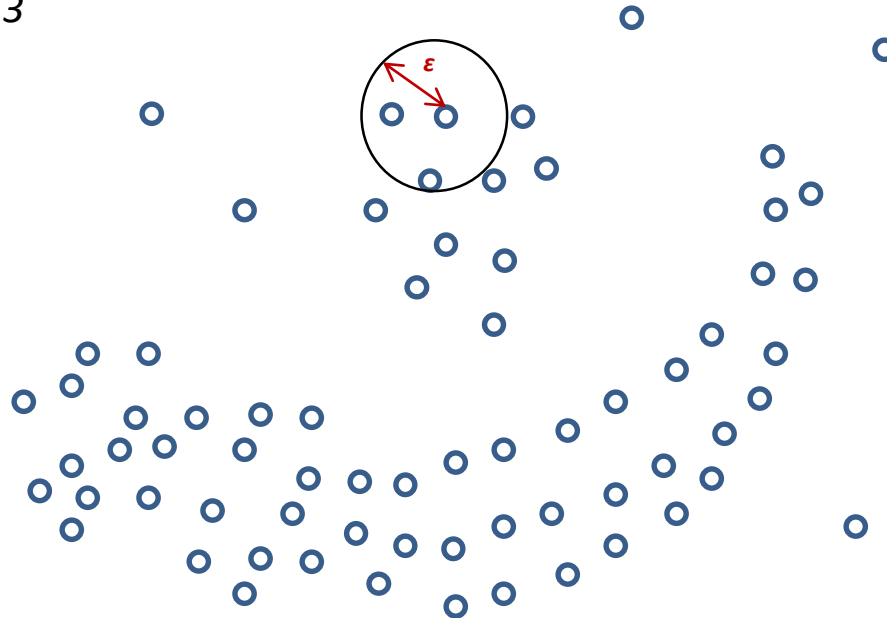
 1. **Pick** a **data point** at random
 - That has not been assigned to a cluster or been designated as an *outlier*
 2. **Compute** its **neighborhood** to determine if it's a *core point*
 - If yes, start a cluster around this point. If no, label the point as an *outlier*
 3. **Expand clusters** by adding all **reachable** points to the cluster
 4. Repeat step 3 until all points are either:
 - Assigned to a cluster or
 - Designated as *outliers*

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5. Clustering Algorithms

5.3. Other Approaches: Density-based Clustering

- Example:
 - ϵ with $MinPts = 3$

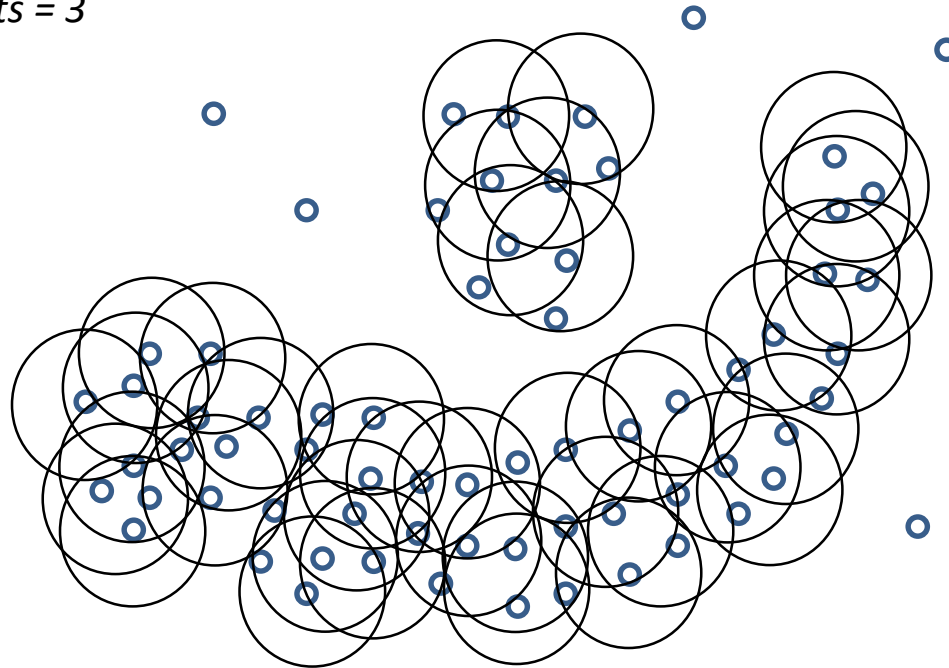


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5. Clustering Algorithms

5.3. Other Approaches: Density-based Clustering

- Example:
 - ε with $MinPts = 3$

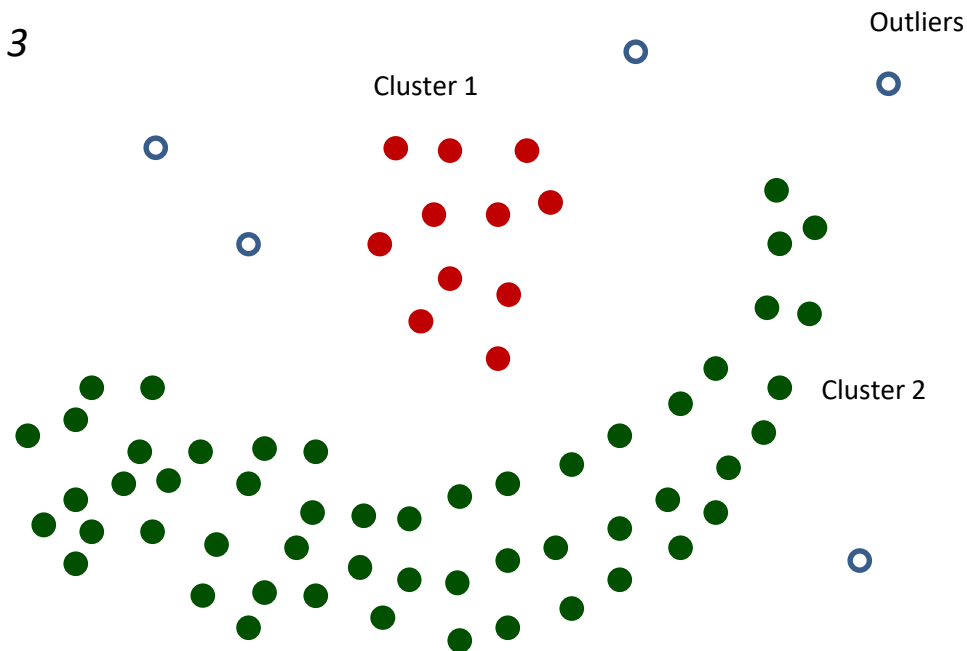


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5. Clustering Algorithms

5.3. Other Approaches: Density-based Clustering

- Example:
 - ε with $MinPts = 3$



Ch. 10 – Data Clustering

5. Clustering Algorithms

5.3. Other Approaches: Density-based Clustering

- **Density-based clustering** handles some of the problems of previous approaches
- Pros:
 - Does not require a user-defined number of clusters
 - Can effectively detect **non-globular** shaped clusters
 - Can effectively handle **noise** and **outliers**
- Cons:
 - **User** needs to define **input parameters**: neighborhood radius ϵ and minimum size of neighborhood *minPts*
 - ⇒ Which might affect the quality of the clustering result
 - **Not efficient** computationally: of $O(n^2)$ time and space
 - Tends to **disregard low-density regions** all together which might include relevant data partitions to be considered in the clustering result

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.3. Other Approaches: Fuzzy Clustering

- Using **typical clustering algorithms**: output clusters **are hard**
 - A hard partition of the data into **separate** (orthogonal) **groups** (clusters)
- **Fuzzy** (or soft) **clustering**: producing fuzzy/soft clusters
 - Soft partitions of the data where each data entity (e.g., XML document, fragment, element) has a variable degree of membership in each of the output clusters
- Main algorithm: **Fuzzy C-Means**
 - Adaptation of the **k-means** partitional algorithm

???

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5. Clustering Algorithms

5.3. Other Approaches: Fuzzy Clustering

- **C-means** is one of the most popular fuzzy clustering algorithms
 - Requires the number of clusters **k** to be specified by the user
- Main process:

 1. Initialization: Select **k** data objects (points) as the initial **centroids**
 2. **Repeat**
 3. Membership: Compute membership degree of every objects w.r.t. **k** centroids
 - **Membership degree** is usually evaluated as **similarity**
 1. Update: Re-compute the **centroids** based on membership degrees
 2. **Until** reaching **convergence**

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5. Clustering Algorithms

5.3. Other Approaches: Fuzzy Clustering

- Mathematically:
 - Initialization: Given **k** chosen by user
 - Randomly choose initial **k** centroids (means) at iteration **t₀**: $m_1^{(t_0)}, \dots, m_k^{(t_0)}$
 - Membership: Given the **k** centroids at iteration **t**
 - Compare every object **x_p** with every centroid **m_i**
 - Compute membership of **x_p** with respect to every centroid **m_i**, noted **w(x_p, m_i)**

$$\Rightarrow w(x_p, m_i^{(t)}) = Sim(x_p, m_i^{(t)})$$
, yet other formulas can also be used
 - Update: Re-computing centroids as means of all data objects
 - Weighted via their membership degrees w.r.t. the centroid at hand

$$\Rightarrow m_i^{(t+1)} = \frac{1}{\sum_{\text{all } x_p} w(x_p, m_i^{(t)})} \sum_{\text{all } x_p} w(x_p, m_i^{(t)}) \times x_p$$

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5. Clustering Algorithms

5.3. Other Approaches: Fuzzy Clustering

- Mathematically:
 - Convergence: can be evaluated in different ways
 - Verifying that **object membership degrees** in iteration **t** did not change in iteration **t+1**
 - More that a certain pre-defined **threshold ε**

$$\Rightarrow \sum_{i=1}^k \sum_{\text{all } x_p} \left| w(x_p, m_i^{(t+1)}) - w(x_p, m_i^{(t)}) \right| \leq \varepsilon$$

- Verifying that fuzzy clusters at iteration **t** have **higher (or close enough) intra-cluster similarity** than fuzzy clusters at iteration **t+1**
 - Can be evaluated as **Sum of Squared Error** (or SSE)

$$SSE^{(t)} = \sum_{i=1}^k \sum_{\text{all } x_p} \text{Dist}(x_p, m_i^{(t)})^2 \quad \text{where } \text{Dist} \text{ is the inverse function of } \text{Sim}$$

$\Rightarrow$ **Higher intra-cluster similarity** means **lower SSE**

- User can also choose allowed SSE difference threshold

Ch. 10 – Data Clustering

5. Clustering Algorithms

5.3. Other Approaches: Fuzzy Clustering

- **NOTE:** Notion of **cluster** does **not** truly **exist** with fuzzy clustering
 - We only manipulate **centroids** and **membership degrees**
- Fuzzy algorithms are typically **partitional**
 - ⇒ and thus carry corresponding **pros** and **cons**
- A **fuzzy** clustering **can be converted** to a **hard** clustering
 - By exclusively assigning data to the **centroid**
 - with the **largest membership degree**

Ch. 10 – Data Clustering

6. Clustering Evaluation

- **Cluster evaluation** (of validation) can be done in two ways:
 - **Internal**: evaluation based on the data that was clustered itself
 - Objective with these measures is to detect:
 - **High** similarity within a cluster (**intra-cluster**)
 - and **low** similarity between clusters (**inter-cluster**)
 - **External**: evaluation based on labeled (expert) data
 - Provided as reference: to be compared with the machine produced clusters

Ch. 10 – Data Clustering

6. Clustering Evaluation

- Internal evaluating measures:

- **Davies–Bouldin index:**
$$DB = \frac{1}{k} \sum_{i=1}^k \max_{i \neq j} \left(\frac{\sigma_i + \sigma_j}{Dist(m_i, m_j)} \right)$$

- Where σ_i is the average distance of all data objects in cluster C_i w.r.t. centroid m_i

$$\Rightarrow \sigma_i = \arg \max_{x_p \in C_i} Dist(x_p, m_i)$$

$\Rightarrow$ **Smallest DB index** designates **best clustering**

- **Dunn index:**
$$Dunn = \frac{\min_{1 \leq i < j \leq k} InterDist(C_i, C_j)}{\max_{1 \leq r \leq k} IntraDist(C_r)}$$

$\Rightarrow$ Identifying **dense** and **well-separated** clusters

Ch. 10 – Data Clustering

6. Clustering Evaluation

- **External evaluating measures:**
 - **Precision** and **Recall**: percentage of **relevant documents** correctly **clustered** by the system with respect to user defined **clusters**
 - For every produced cluster C_i , consider:
 - a_i : # of data objects in C_i that indeed correspond to C_i (correctly clustered)
 - b_i : # of data objects in C_i that do not correspond to C_i (miss-clustered)
 - c_i : # of data objects not in C_i , although they correspond to C_i (documents that should have been clustered in C_i)

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6. Clustering Evaluation

- **External evaluating measures:**

- **Precision:**
$$PR = \frac{\sum_{i=1}^n a_i}{\sum_{i=1}^n a_i + \sum_{i=1}^n b_i} \in [0, 1]$$

- **High precision** denotes that the clustering task achieved high accuracy
 - Grouping together objects that should actually be grouped together

- **Recall:**
$$R = \frac{\sum_{i=1}^n a_i}{\sum_{i=1}^n a_i + \sum_{i=1}^n c_i} \in [0, 1]$$

- **High recall** means that very few objects are not in the appropriate cluster

Ch. 10 – Data Clustering

6. Clustering Evaluation

- **External evaluating measures:**
- In addition to comparing one approach's **precision** with another's **recall**
 - It is a common practice to consider the **F-value**
 - which represents the **harmonic mean** of *precision* and *recall*

$$\text{F-value: } F\text{-value} = \frac{2 \times PR \times R}{PR + R} \in [0, 1]$$

- **High *precision*** and ***recall***, and thus **high *F-value*** indicate **high clustering quality**
- **NOTE:** **clustering quality** depends not only on the clustering algorithm
 - But also (and sometimes mostly) on the **features** and the **similarity measure** used

Ch. 10 – Data Clustering

Wrapping up

- Clustering consists in finding groups of data objects such that
 - Objects in a group will be **similar** (or related) to one another
 - And **different** from (or unrelated to) objects from other groups
- Clustering is also known as **un-supervised learning**
 - Versus **supervised learning** where the groups of data are known
- Various clustering approaches and algorithms exist
 - Partitional, hierarchical, incremental, constrained agglomerative, spectral, density-based, and fuzzy clustering, among others
 - Each with its **pros** and **cons**
- Various measures also exist for **evaluating clustering** quality
 - **Internal** such as the **Dunn index**, and **external** such as **Precision** and **Recall**

Ch. 10 – Data Clustering

Questions?



Ch. 10 – Data Clustering

Exercises and Tutorials

- Provided by the instructor during lecture sessions



2nd project assignment



Ch. 10 – Data Clustering

Main References

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